

XAI and FAIR Data in Insurance: A Systematic Review

Kemmy Business School, University of Limerick

MSc Risk Management & Insurance

Emer Owens

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This dissertation is solely the work of the author and submitted in partial fulfilment of the requirements of the MSc Risk Management and Insurance.

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Abstract

Artificial Intelligence (AI) is posited to transform the insurance industry with AI models' capability to perform highly complex computational tasks. However, transparency in AI models is pertinent in the industry due to the highly sensitive data inputted into the models, with 'black-box' AI models posing problems. Explainable Artificial Intelligence (XAI)'s abilities set to reintroduce understandability and transparency in AI models. The paper presents a systematic review of literature on the ubiquity of XAI methods within the insurance industry. Through the review and classification of the literature within the framework of the Insurance Value Chain (IVC), explainable systems' current state-of-play is discussed. Based on a dataset of 95 articles (2000-2021) on the current use of AI in insurance and risk management, the author highlights the prevalence of XAI methods on the IVC's various stages. The research findings suggest that XAI methods are particularly useful in Claim Management and Underwriting & Pricing practices, while the primary XAI technique used within the IVC is Knowledge Distillation & Rule Extraction. To the best of the author's knowledge this is the only study on XAI's current state-of-play within the insurance industry, while simultaneously contributing to the wider research space on XAI. The paper also addresses the FAIR Data Principles' (Findable, Accessible, Interoperable, Reusable) prevalence in insurance AI research. The author highlights lacking research reproducibility within the reviewed articles which hinders further XAI and insurance research. The development of a framework to anonymise insurance data for research use is recommended, with the aim of preserving insureds' data anonymity and maintain trust in AI models used by insurance companies. An additional contribution of the paper is the proposal of a revised XAI definition for use within the financial services and insurance industries, which is informed by the systematic review of AI applications on the IVC and previous definitions of XAI in the literature.

Keywords

Explainable Artificial Intelligence, Machine Learning, Insurance Value Chain, FAIR Data Principles

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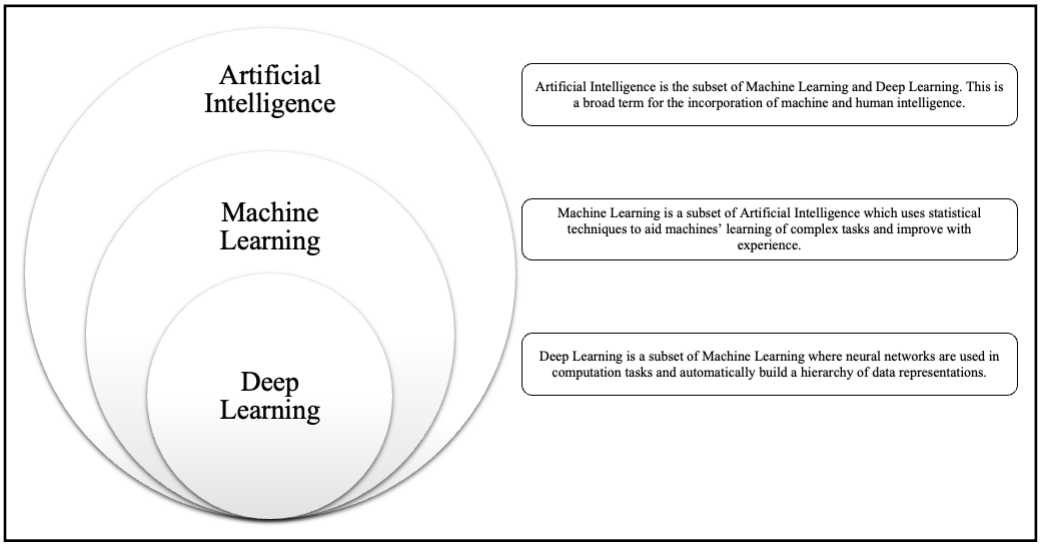
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1. Introduction

Artificial Intelligence (AI) revenues in insurance are expected to grow 23% to $3.4 billion between 2019-2024, yet the suitability of black-box AI models in insurance practices is questionable (Bean 2021; GlobalData 2021). The growth of AI as intelligent machines which hold the potential to perform complex computational tasks is revolutionising financial services, particularly within the insurance industry. Data and its potential use is seen as a primary strategic asset and a source of competitive advantage in financial services firms, with AI models' leverage of such data providing numerous advantages (Kim and Gardner 2015). Such advantages of AI use in the insurance industry include enhanced fraud detection in claims management, additional granularity when pricing insurance premiums, the creation of smart contracts and analysis of legal documents. AI encompasses the collation of multiple technologies in a single system which enables machines to interpret data and aid complex computational decision-making (Chi *et al.* 2020). Although AI models' advantages abound, recent literature highlights the models' decision-making processes' opacity – primarily the lack of transparency and explainability – which is coined as black-box thinking (Adadi and Berrada 2018). There is potential for these models' impediments to compound and extenuate bias in their decision-making processes with unfair outcomes possible within the insurance industry. The Insurance Value Chain (IVC) makes extensive use of AI methods at every stage of the value creation process, with AI particularly impactful in claims management and underwriting & pricing departments (Eling *et al.* 2021).

The mushrooming growth of research in Explainable Artificial Intelligence (XAI) since 2010 provides a remedy for the abovementioned AI pitfalls (Vilone and Longo 2020). Throughout this systematic review, AI is used as an umbrella term for both Machine Learning (ML) and Deep Learning (DL) (*Figure 1*). As an extension of AI models, XAI involves enhancing current AI models by developing their transparency and interpretability, with such AI advancements ultimately aiming to make AI models more understandable to humans (Adadi and Berrada 2018). XAI methods promise to enhance the high-performing abilities of AI to learn and model complex tasks, while also reintroducing the human brain as a key decision-maker. The understanding of machine's internal mechanisms and the ability to alter contentious decision-making pathways allows for the human brain to trust in machines again (Koene *et al.* 2019).

Figure 1: The Umbrella of Artificial Intelligence. Artificial Intelligence encompasses both Machine Learning and Deep Learning models.



The rationale for a systematic review of literature on XAI, with specific consideration paid to its use in risk analysis and insurance industry practices is pertinent. This is due to its expanding impact on the financial services industry at large and the inevitable development of refined regulatory considerations specific to the field of XAI. A collation of relevant academic and professional literature on XAI will be useful for regulators and insurance practitioners in the coming years to aid informed decision-making surrounding ethical considerations of XAI and insurance regulation. Regulatory considerations, risk analysis, and the inexorable impact on key insurance dynamics provide definitive reasoning for an in-depth overview of XAI's use within the insurance industry.

Another key rationale for the importance of explainable and interpretable AI methods in insurance is the societal impact of insurance provision and affordability (Daniels 2011). Insurance affordability is a key determinant of societal progress, with the modelling of insurance pricing practices playing a key role in this affordability. The actuarially fair pricing of insurance premiums allows for a population to access insurance at rates which they can reasonably afford (Grant 2012). When an AI model is tasked with this complex pricing task, the ability for humans to discern the inner-workings of the model to ensure the premium rate which the AI model outputs is actuarially fair is pertinent to the continuation of societal progress. Albeit a traditionally reactive, rather than proactive industry to technological advancements (Agababa 2017), the adoption of XAI methods

in insurance practices, especially in Underwriting and Pricing, is a sound rationale for the promotion of explainable and interpretable AI models in the insurance industry.

The primary contribution of this systematic review to the font of XAI knowledge is an in-depth analysis of published literature on AI in insurance practices[1] while assessing these AI methods based on their degree of explainability using defined XAI criteria. This review of key literature provides a comprehensive analysis of XAI applications in insurance for both key insurance regulators and insurance practitioners which will allow for extensive application in future regulatory decision-making. This paper provides a synthesis of current understandings of XAI methods in various industry practices to inform the overall font of XAI knowledge. Contradictions and comparisons are drawn between proposed definitions and taxonomies of XAI in literature.

In line with a discussion of transparency in the insurance industry, this paper also provides the significant contribution of a primary analysis of study reproducibility in insurance research. The FAIR Data Principles act in defining key criteria for reproducible studies across a wide range of research disciplines. To create an academic scholarship of open data sources, the FAIR principles outline guidelines for researchers on how to 'FAIRify' their research datasets. This systematic review critically assesses the reviewed articles based on defined criteria for reproducible datasets, highlighting a myriad of irreproducible studies. The author outlines the challenges concerning open data sources in the insurance industry, citing sensitive information transactions and insureds' trust in their insurance company as key obstacles. To rectify the reproducibility issue in insurance research, a framework for the anonymisation of insurance data for research purposes is proposed to aid the expanding network of insurance research across the entire IVC.

This paper is structured as follows; Section 2 presents related works of this review while analysing current research on XAI's definition and related taxonomies. This section also provides related work on AI's impact on the IVC and a background to the FAIR Data Principles. Section 3 outlines the methodological system to collect and analyse relevant literature on AI use along the IVC. The search technique to arrive at relevant articles is especially emphasised to ensure the validity of eventual research results and allow for future research reproducibility. Section 4 outlines the review's findings of the systematically chosen sample of literature and their AI methods through

[1] Including both life and non-life insurance activities.

the lens of defined XAI criteria and the assessment of each article in line with the FAIR Data Principles guidelines. Section 5 presents a novel discussion of the review's results on the ubiquity of XAI along the IVC, focusing on the extent to which AI applications along the IVC are explainable. This section also discusses the reproducibility of the reviewed research as defined by the FAIR Data Principles. Section 6 concludes the systematic review, reiterating points of interest regarding the future of XAI applications and FAIR publicly available datasets in insurance practices. Also provided are comments on further research areas on XAI in the insurance industry and the 'FAIRification' process of insurance research datasets.

1.1 Key XAI Terminology

Before reviewing research in the field of XAI and AI applications in insurance, it is necessary to define some key terminology used throughout the review paper. The review is centred on the classification of XAI systems, therefore certain criteria[2] which determine the explainability of AI models are used throughout the review as summarised below.

Kelley *et al.* (2018) define AI as 'a computer system that can sense its environment, comprehend, learn, and take action from what it's learning', with XAI intuitively expanding on this description by allowing humans to be present at every stage of this AI decision-making lifespan. A common misnomer of AI models' explainability is that it is simply the improvement of trust in AI systems and their decision processes, through developing 'causal structures in observational data' (Goodman and Flaxman 2017; Lipton 2018). In actual fact, models' explainability enhances their interpretability, i.e. understanding *how* a model came to a certain decision (Lou *et al.* 2013), while also positively impacting fair and ethical decision making for high-computational tasks (Srihari 2020). The explainability and interpretability of models used in insurance industry practices will prove of prime importance when discerning appropriate regulatory measures for XAI systems' use in the future.

[2] Primarily adapted from Payrovnaziri *et al.* (2020)'s systematic research on XAI models and Electronic Health Records in medicine.

1.1.1 XAI Variables

Key XAI variables and criteria used both in the systematic review and throughout this paper are briefly outlined below as a precursor to later findings and discussion. These XAI groupings are derived from Payrovnaziri *et al.* (2020), who synthesised the groupings from Du *et al.* (2019) and Carvalho *et al.* (2019)'s XAI reviews.

Intrinsic vs. Post-hoc Interpretability

The main differentiating aspect between an intrinsic and a post-hoc interpretable explanation is whether interpretability is achieved through imposing constraints on the complexity of the model (intrinsic) or whether the models' explainability was analysed after training (post-hoc) (Molnar 2019). Intrinsic methods primarily describe *how the model works*, which denotes a high degree of transparency in the model which is interpretable by itself (Lipton 2018; Rudin 2018). Lipton (2018) contrastingly summarises post-hoc explainability as *what else can the model tell us*. Carvalho *et al.* (2019) clarifies that it is possible to apply post-hoc methods to intrinsic models, as post-hoc methods are usually derived from the main model. In summary, intrinsic models achieve their interpretability incorporating it directly into their structures, while post-hoc models require the creation of a second model to provide explanations for the existing model (Du *et al.* 2019).

Local vs. Global Interpretability

Local explanations primarily reveal the impact of input features on the overall model, while local explanations are combined to reach a global explanation in an XAI model. Popular local explanation methods include: 1) the reporting of the decision path, 2) the assigning of credit to each input feature in the model and, 3) the application of several model-agnostic approaches which require the repeated execution of the model for each explanation (Baehrens *et al.* 2010; Štrumbelj and Kononenko 2014; Lundberg *et al.* 2020). A global explanation provides an overall view of the AI system, through listing the system's rules or features that eventually determine their outcome (Lundberg *et al.* 2020). In terms of trustworthiness, Adadi and Berrada (2018) cite local

explanations as more trustworthy than global ones as the latter connotes a sense of understanding of the mechanism by which the model works.

Model-specific vs. Model-agnostic Interpretation

Both model-specific and model-agnostic interpretation methods are derived from the above intrinsic vs. post-hoc explainability criterion. As the name suggests, model-specific interpretation methods are limited to specific model classes as each method is based on a specific model's internals (Molnar 2019). Model-specific interpretability is by definition achieved from Intrinsically Interpretable Models (Adadi and Berrada 2018; Carvalho *et al.* 2019). Alternatively, model-agnostic methods can be applied to any model (black-box or otherwise) and are applied after the model has been trained (similar to post-hoc interpretability). This method includes the analysis of relationships between the system's feature inputs and outputs, without sacrificing the model's predictive power (Lipton 2018; Carvalho *et al.* 2019). *Table* 1 provides a summary of the above interpretability criteria and their generalised relationships.

Table 1: Association Between XAI Interpretability Criteria

In-model	Intrinsic	Model-specific
Post-model	Post-hoc	Model-agnostic

1.1.2 XAI Categories

The following categories of XAI methods are used to classify published applications of AI in insurance: 1) Feature Interaction and Importance, 2) Attention Mechanism, 3) Dimensionality Reduction, 4) Knowledge Distillation and Rule Extraction, and 5) Intrinsically Interpretable Models. Each of these XAI categories are identified by Payrovnaziri *et al.* (2020) as essential for the assessment of AI methods' degree of explainability. The categories are discussed further in *Section 4.2* in conjunction with their specific use along the IVC.

2. Background and Related Works

Relevant related work for this systematic review is partitioned into three sections: 1) studies on XAI, 2) studies on AI in insurance and, 3) studies on the FAIR Data Principles. This section highlights the gap in the literature on XAI in the insurance industry and defines the unique contribution of this paper to the research field on XAI.

2.1 Explainable Artificial Intelligence

Related works for this paper involve the vast body of literature which survey and review XAI methods and classify the field of explainable artificial models in various research areas. Gade *et al.* (2019) outline the main challenges for XAI researchers which include 1) 'defining model explainability', 2) 'formulating explainability tasks for understanding model behaviour and developing solutions for these tasks', and 3) 'designing measures for evaluating the performance of models in explainability tasks'. Vilone and Longo (2020) later systematic study contributed a classification system for published XAI literature, aiming to establish boundaries in the field of XAI research. Four main clusters of research were found by Vilone and Longo (2020); 1) 'reviews focused on specific aspects of XAI', 2) 'the theories and notions related to the concept of explainability', 3) 'the methods aimed at explaining the inferential process of data-driven and knowledge-based modelling approaches', and 4) 'the ways to evaluate the methods for explainability' (Vilone and Longo 2020).

Extending on the above, *Table* 2 provides a summary of other research on XAI, highlighting the ongoing efforts in academic literature to develop a sound and thorough definition of XAI. The definition of XAI developed in each paper is highlighted, which is commonly referred to as 'explainability' rather than 'interpretability', as Islam *et al.* (2020) notes that explainability is more than interpretability in terms of importance and trust in the prediction. Interpretability is often the end-goal, with explanations acting as tools to reach interpretability (Honegger 2018). Additionally, the GDPR which is discussed later in this paper covers only explainability (Došilović *et al.* 2018). Instead of offering actionable definitions of XAI, other works classify the requirements that an explainable system should meet (Lipton 2018; Xie *et al.* 2020) or the methods of evaluations under which an AI system can be deemed explainable (Doshi-Velez and Kim 2017; Lipton 2018).

Table 2: XAI definitions found in the literature. Note that these definitions refer to the notion of "explainability", which is often confused with "interpretability". "Explainability" as a foundational definition for XAI is most common throughout the literature.

XAI Review Paper	Reference	XAI Definition
A unified approach to interpreting model predictions	Lundberg and Lee (2017)	Explainability as the "interpretable approximation of the original complex model"
Explainable software analytics	Dam *et al.* (2018)	Explainability "measures the degree to which a human observer can understand the reasons behind a decision (e.g., a prediction) made by the model"
A Survey of Methods for Explaining Black Box Models	Guidotti *et al.* (2018)	Explanation methods are referred to as one of the following statements: (i) explaining the model, (ii) explaining the outcome, (iii) ability to inspect the internals of a black-box model or (iv) providing a transparent solution to the prediction problem
Hype cycle for artificial intelligence	Sicular *et al.* (2018)	"Explainable AI is a set of capabilities that describes a model, highlights its strengths and weaknesses, predicts its likely behaviour, and identifies any potential biases. It can articulate the decisions of a descriptive, predictive, or prescriptive model to enable accuracy, fairness, accountability, stability, and transparency in algorithmic decision making"
Methods for interpreting and understanding deep neural networks	Montavon *et al.* (2018)	Explanation as the "collection of features of the interpretable domain, that have contributed for a given example to produce a decision (e.g., classification or regression)"
Explainable agents and robots: Results from a systematic literature review	Anjomshoae *et al.* (2019)	XAI is an "artificial intelligence whose actions, recommendations and the underlying causes to its decisions are understandable by humans"
Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy	Dwivedi *et al.* (2019)	"Explainability is the ability to explain the reasoning behind a particular decision, classification or forecast"
Explanation in artificial intelligence: Insights from the social sciences	Miller (2019)	"XAI refers to an explanatory agent revealing underlying cause to its or another agents' decision-making"
Contextual Explanation Networks	Al-Shedivat *et al.* (2020)	Explainability is a "local approximation of a complex model (by another model)"
Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities, and challenges toward responsible AI	Arrieta *et al.* (2020)	"Given an audience, an explainable AI is one that produces details or reasons to make the functioning clear or easy to understand"
Opportunities and challenges in explainable artificial intelligence (XAI): A survey	Das and Rad (2020)	"A suite of techniques and algorithms designed to improve the trustworthiness and transparency of AI systems"
Towards quantification of explainability in explainable artificial intelligence methods	Islam *et al.* (2020)	"Explainability of an AI model's prediction is the extent of transferable qualitative understanding of the relationship between model input and prediction (i.e., selective/suitable causes of the event) in a recipient friendly manner"
Explainable Artificial Intelligence: A Systematic Review	Vilone and Longo (2020)	Explainability is "the collection of features of an interpretable domain that contributed to produce a prediction for a given item"
XAI in the Financial Sector	van den Berg and Kuiper (2020)	"Given a stakeholder, XAI is a set of capabilities that produces an explanation (in the form of details, reasons, or underlying causes) to make the functioning and/or results of an AI solution sufficiently clear so that it is understandable to that stakeholder and addresses the stakeholder's concerns"
XAI Handbook: Towards a Unified Framework for Explainable AI	Palacio *et al.* (2021)	"An explanation is the process of describing one or more facts, such that it facilitates the understanding of aspects related to said facts (by a human consumer)"
Explainable Artificial Intelligence (XAI)	Turek (n.d)	XAI aims to "produce more explainable models, while maintaining high levels of learner performance (prediction accuracy); and enable human users to understand, appropriately trust and, effectively manage the emerging generation of artificially intelligent partners"

Reviews of XAI in medicine ignited the XAI research field, with many studies on the technology's effects on disease diagnosis, classification and treatment published in recent years. Payrovnaziri *et al.* (2020) involved the review of 49 articles published in the period 2009 – 2019 to group XAI methods used in the medical field. In this study, Payrovnaziri *et al.* (2020) grouped XAI methods into 5 different groups: 1) 'Knowledge Distillation and Rule Extraction', 2) 'Intrinsically Interpretable Models', 3) 'Data Dimensionality Reduction', 4) 'Attention Mechanism' and 5) 'Feature Interaction and Importance'. Mathews (2019) performs an evaluation of XAI methods on biomedical signal classification for cancer detection, while Sarp *et al.* (2021) classified chronic wounds using XAI methods. Antoniadi *et al.* (2021) outline challenges pertaining to AI's use for clinical decision support systems, emphasising lacking transparency as a key issue. Notwithstanding the obvious advantages of XAI methods to enhance understandability and aid medical practitioners' decisions abound, their research finds a distinct lack of XAI applications in medicine.

Finance-related studies on XAI include Demajo *et al.* (2020), Hadji Misheva *et al.* (2021) and Biecek *et al.* (2021)'s research on credit scoring and risk management. Similarly, Bussmann *et al.* (2020) explore XAI in fintech risk management and peer-to-peer lending platforms, while Kute *et al.* (2021) also focus on risk management in finance applications through their review of DL and XAI technologies in the identification of suspicious money laundering practices. Adadi and Berrada (2018) provide a foundational background to the main concepts and implications of an XAI system, citing data security and fair lending in financial services as key issues surrounding XAI use in financial services. With respect to banking and accounting practices, Burgt (2020) states that trust of AI systems in the banking industry is paramount and provides a discussion on the trade-off between explainability and predictability of AI systems. Gramespacher and Posth (2021) then utilise XAI to optimise the return target function of a loan portfolio, while Mehdiyev *et al.* (2021) add to the conversation through analysing tax auditing practices and public administration's appetite for XAI. Albeit the obvious advantages of developing transparent decision-making systems in public administration, this research cites the requirements of safe, reliable, and trustworthy AI systems as creating additional complexity in AI systems which take some time to implement widely.

The interest in human-centred decision-making machines reaches beyond medical and finance domains. Putnam and Conati (2019) provide a survey that finds students seek additional explanations from their Intelligent Tutoring System to aid their education prospects. Natural Language Processing (NLP) is another research area with significant interest in XAI methods as revealed by Danilevsky *et al.* (2020), with sarcasm detection in dialogues later reviewed by Kumar *et al.* (2021). Anjomshoae *et al.* (2019) reviews inter-robot explainability and addresses the issue of explainability to non-users of ML robots through personalisation and context awareness.

The current systematic review will build upon previous research on XAI methods' classification and analysis of XAI literature during the systematic selection of literature. Although the above literature does provide a brief overview of the current understanding of XAI and related key concerns highlighted in the literature, this is the first paper to review XAI applications in the insurance industry.

2.2 Artificial Intelligence Applications in Insurance

Tekaya *et al.* (2020) preface AI research in financial services through offering an overview of the current use-cases and advantages of implementing Big Data and AI models in banking, credit risk management, fraud detection and the insurance industry. Several other articles highlight the importance and advantages of AI applications in the insurance industry, predicting major shifts in operations in the coming years (Umamaheswari and Janakiraman (2014); (Riikkinen *et al.* 2018; Paruchuri 2020)). Popular areas within insurance research where AI has been applied include fraud detection (Sithic and Balasubramanian (2013) and Verma *et al.* (2017)) and claims reserving (Wüthrich (2018), Baudry and Robert (2019), Blier-Wong *et al.* (2021) and Lopez and Milhaud (2021)). Grize *et al.* (2020) focus on ML applications in non-life insurance, highlighting AI's positive impact on risk assessment to improve the insurance companies' overall profitability in the long run.

Fang *et al.* (2016) used Big Data to develop a new profitability method for insurers using historical customer data, where they found that the Random Forest (RF) model outperformed other methods of forecasting (linear regression and SVM). Shapiro (2007) documents the extent to which fuzzy

logic (FL) has been applied to insurance practices, which prompted Baser and Apaydin (2010)'s later research on claims reserving using hybrid fuzzy least squares regression and Khuong and Tuan (2016)'s creation of a neuro-fuzzy inference system for insurance forecasting. NallamReddy *et al.* (2014) present a robust review of clustering techniques used in insurance. Quan and Valdez (2018) use another understandable and transparent AI method, Decision Trees (DT), to investigate their use in insurance claims prediction. Interestingly, later research acknowledges the low predictive power of DTs and boosts their intrinsic interpretability to provide a more robust insurance pricing model (Henckaerts *et al.* 2021).

Sarkar (2020) argues that the insurance industry holds the potential for algorithmic capabilities to enhance each stage of the industry's value chain. Through highlighting AI's offerings at each stage of the IVC, the research prompted further studies from Walsh and Taylor (2020) and Eling *et al.* (2021) to determine precise AI opportunities available to the insurance industry. Walsh and Taylor (2020) highlight AI models' ability to mimic, or augment, human capabilities with NLP, Internet of Things (IoT) and computer vision. Eling *et al.* (2021) analyse AI's impact at each step on IVC and specifically highlights the potential for AI to enhance revenue streams, loss prediction and loss prevention measures for insurance practitioners.

2.3 Reproducibility and FAIR Data in AI Literature

Throughout each stage of a research project, appropriate data management is essential for research validation. Akin to the ugly duckling in research, proper data management is a step researchers understand they *ought* to engage in, but rarely want to (Kowolenko and Vouk 2018). Reasons abound for the necessity of reproducible research findings, including the opportunity to validate research findings and encourage the reusability of data and research methodologies to further contribute to a given research space (Madeyski and Kitchenham 2015; Peng and Hicks 2021). Issues with data management arise as metadata capturing is not implemented in everyday scientific research practices. This is owed in part to lacking data management training and the general understanding of data management occurring at the end of a project, after the data is published (Papadiamantis *et al.* 2020; Furxhi *et al.* 2021).

The establishment of the FAIR data principles in 2014 (published in 2016[3]) heralded the beginning of inter- and cross-disciplinary research transparency through the establishment of a minimal set of rules for research data stewardship in the life sciences domain (Boeckhout *et al.* 2018). Wise *et al.* (2019) emphasise the importance of research data which complies with the FAIR guidelines: Findable, Accessible, Interoperable and Reusable data enable digital transformation and societal progress (Romanos *et al.* 2019). A significant note on the principles is that they precede the implementation choices of models, therefore only provide a set of recommendations for data researchers and publishers to assist the evaluation of the data being used (Wilkinson *et al.* 2016). Although a step in the right direction for research transparency, there lacks actionable steps in the implementation of FAIR data practices in many research areas, with the risk of incompatible implementations likely to cause future issues for researchers (Jacobsen *et al.* 2020).

Both governments and researchers are exploring alternative methods of data governance as a means of increasing competition in industries particularly impacted by AI, including finance, insurance and medicine (Brodsky and Oakes 2017). Furxhi *et al.* (2021) provide seminal work on the 'FAIRification' process of research datasets and methods in nanotechnology, while Rashid *et al.* (2020) provide a practical approach for proper annotation of datasets which are standardised and machine-readable. Gajbe *et al.* (2021) provide a review of several data management plan tools to assist researchers in choosing appropriate structures to aid their data management process. There appears a shift from viewing datasets as a by-product of a research project to a value-enhancing asset whose value increases the more they are used (Leonelli 2016; Wilkinson *et al.* 2016). This shift necessitates further classification of 'FAIRified' datasets in AI research and the identification of methods to promote the FAIR Principles' implementation in across a wide range of research disciplines.

[3] Consulted and first published under Force11 in 2016.

3. Review Method

3.1 Literature Search Process

This literature search plan and related inclusion and exclusion criterion follow the framework found within Eling *et al.* (2021), with the aim of expanding upon their research to assess the ubiquity of XAI methods in the IVC's AI applications. Eling *et al.* (2021)'s research assessed AI's impact on the IVC and the insurability of risks. The research presented in this paper expands on the abovementioned research to determine not only the impact on the IVC of AI systems being used, but their degree of explainability. The presence of FAIR research along the IVC is also analysed using defined FAIR research criteria.

To examine the current state of literature, the systematic search was conducted on EBSCOhost (Business Source Complete and EconLit), ACM Digital Library[4], and IEEE Xplore. These databases were chosen due to their wide breadth of content spanning both insurance and finance-related research, while also accounting for computer science journals to access research on AI. The above databases were chosen to feasibly and approximately align the current review with Eling *et al.* (2021)'s research, while considering database accessibility limitations. Section 3.3 outlines limitations of the current review, including the author's limited access to the same databases as Eling *et al.* (2021)'s research, highlighting the substitute accessible databases used for this review's literature search process.

Table 3 outlines the key search terms used interchangeably with AI in these databases, alongside 'Insurance' OR 'Insurer' using Boolean terminology. This broad set of search terms ensures an all-encompassing article-base of the IVC's use of AI and are adapted from Eling *et al.* (2021)'s literature search method.

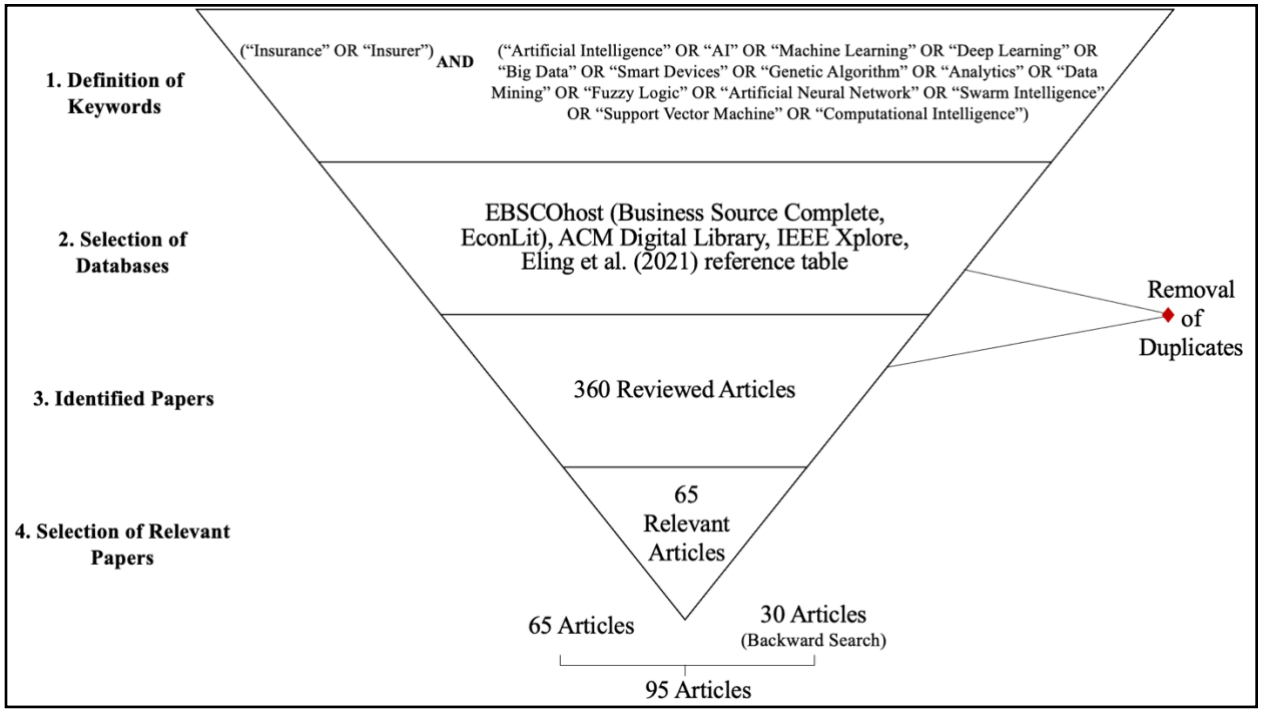
[4] Searched 'The ACM Guide to Computing Literature'

Table 3: Key Search Terms Interchangeable with Artificial Intelligence in the Literature Search Process.

Artificial Intelligence	Smart Devices	Artificial Neural Network
AI	Genetic Algorithm	Swarm Intelligence
Machine Learning	Analytics	Support Vector Machine
Deep Learning	Data Mining	Computational Intelligence
Big Data	Fuzzy Logic	

Figure 2 outlines the systematic literature search process where an initial 360 articles were scanned for relevancy to this paper. Key relevancy criteria included the assessment of articles' contents in relation to their place along the IVC. The IVC criteria is extensively outlined in *Table 4*, as adapted from both Eling *et al.* (2021) and EIOPA (2021)'s research. The articles included in the systematic study of XAI in insurance are categorised according to the specific stage of the IVC which they refer to. This categorisation allows for further assessment of XAI use within the entire IVC process.

Figure 2: Literature Search Process. Backward Searching includes the assessment of the references in each of the 65 relevant articles for additional articles of relevance to the current review.

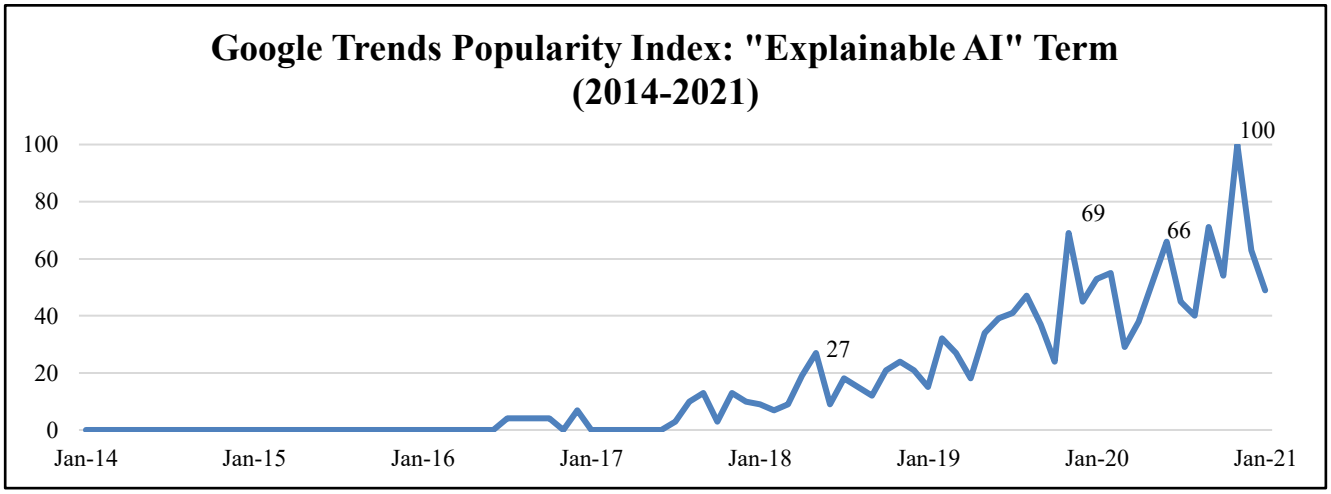


In addition to the above, the articles' relevancy was filtered using the following criteria set:

- *Time Period:* Articles[5] published between 2000-2021 are included,
- *Relevancy:* The presence of keywords (*Table 3*) in the abstract is necessary for the article's inclusion. Additionally, the articles need to be relevant to the assessment of AI applications along the IVC directly (e.g., articles concerned with determining drivers' behaviour using telematics information, which might later inform insurance companies' pricing practices were excluded, as well as generalised surveys on AI uses in insurance[6]),
- *Singularity:* Duplicate articles found across the various databases are excluded,
- *Accessibility:* Only articles which are accessible through the aforementioned databases and are accessible in full text are included (i.e., extended abstracts were not included),
- *Language:* Only articles published in English are included.

Articles published prior to 2000 are not included in this due to the increased understanding of AI from 2000 onwards (Liao *et al.* 2012), and the creation of the General Data Protection Regulation (GDPR) in 2016 (implemented in the European Union in 2018) which will be especially applicable to conversation on future XAI regulation.

Figure 3: The Google Trends Popularity Index analyses the popularity of key search terms in google web searches. A Google Trends Popularity value of 100 is the peak popularity of the term, with 50 denoting the term is half as popular as its potential.

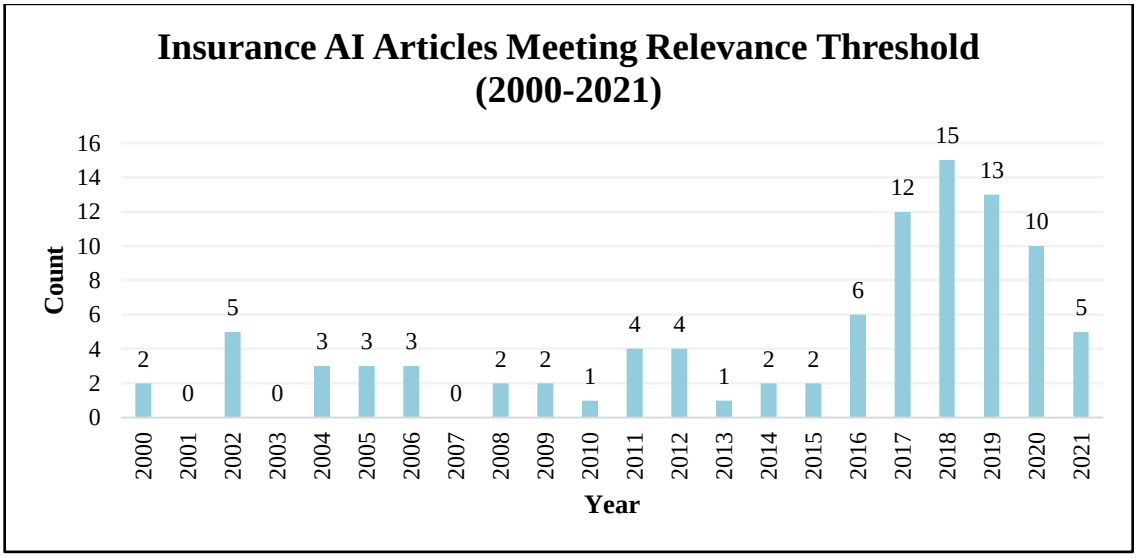


[5] 'Articles' throughout this review refers to both academic articles and conference papers.
[6] Several such surveys and reviews are discussed in *Section 2.2*.

Table 4: The Insurance Value Chain. The stages of the insurance industry's IVC is adapted from Grize *et al.* (2020), Eling *et al.* (2021) and EIOPA (2021).

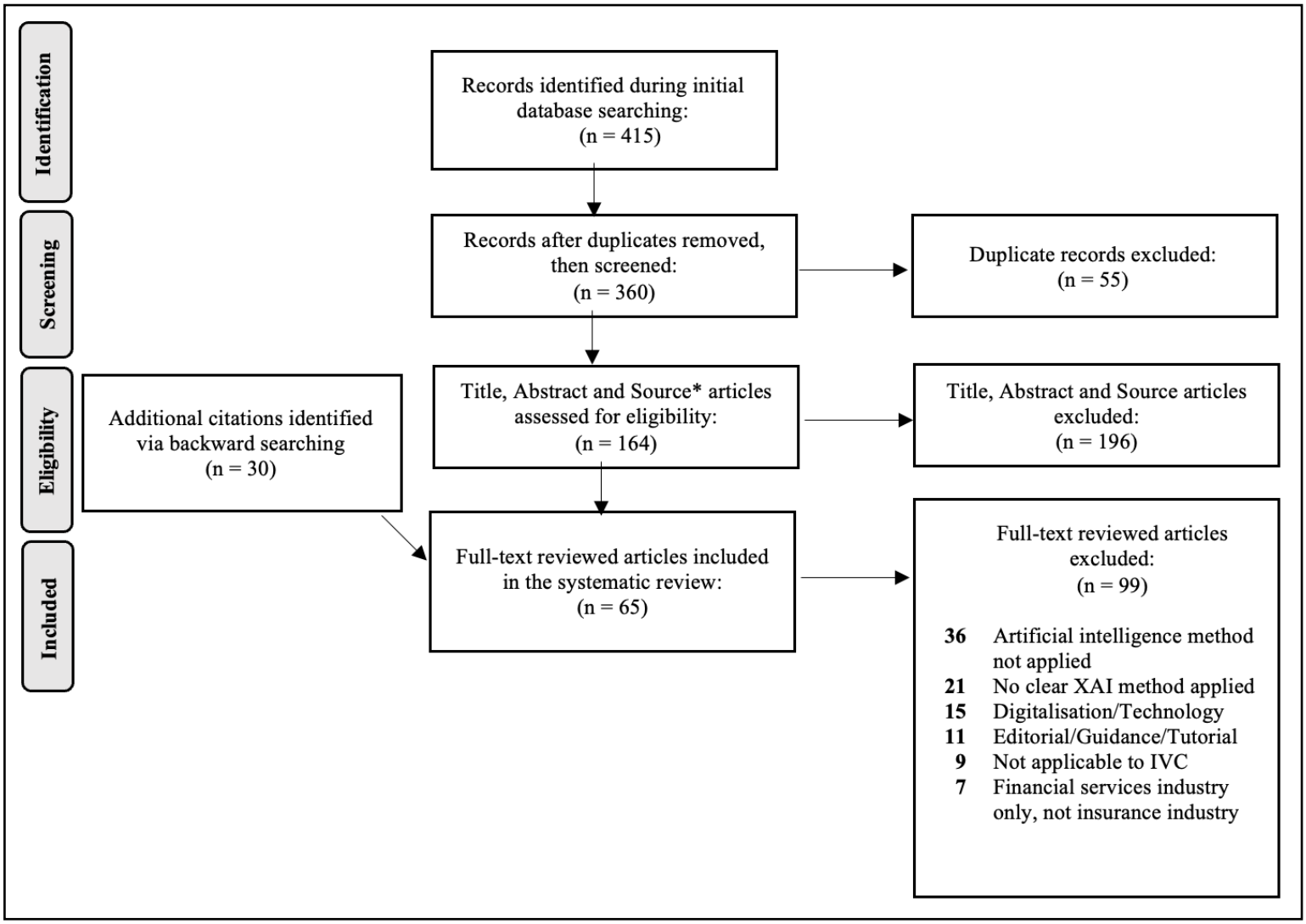
Value Chain Stage	Main Tasks	Impact of Artificial Intelligence Applications
Marketing	Market and customer research Analysis of target groups Development of pricing strategies Design of advertisement and communication	- Improved prediction of customer lifetime value - Enhanced customer segmentation for personalised customer outreach and tailored communication strategies - Advanced insight about preferences in consumer purchasing behaviour for the identification of target product propositions and the generation of new ideas for product innovation - Churn models to enhance customer retention
Product Development	Configuration of products Verification of legal requirements	- Establishment of add-on services such as early detection of new diseases and their prevention enables the development of new revenue streams in addition to risk coverage - Entry into new markets and development of ecosystems with business partnerships in artificial intelligence-driven markets (e.g., autonomous driving, real-time health, and elderly care with nanobots, natural catastrophe management, smart home ecosystems) - Development of novel products utilising AI methods (e.g., usage-based, situational, and parametric insurance)
Sales and Distribution	Customer acquisition and consultation Sales conversations Product sale After-sales services	- Support of human sales agents by offering advanced sales insights (e.g., cross- and up-selling opportunities) through smart data-driven virtual sales assistants (chatbots) for improved customer consultation and tailored product recommendations - Proactive customer relationship management and improved after-sales services through increased client transparency - Chatbots for automated product consultation and sale of standardised insurance products - Customer Relationship Management (CRM) analytics used to inform nudging and cross-selling of related services ("next-best action")
Underwriting and Pricing	Product pricing (actuarial methods) Application handling Risk assessment Assessment of final contract details	- Automated application handling, underwriting and risk assessment processes enable accurate insurance quotes within minutes - New data and insights allow the formation of small and homogenous risk pools, reduction of adverse selection and moral hazard in risk assessment - Micro-segmentation of insurance customers based on behavioural traits to provide personalised insurance pricing (e.g., dynamic online pricing)
Contract Administration and Customer Services	Change of contract data Customer Queries	- Development of chatbots for automated answering of written and verbal customer queries using Natural Language Processing (NLP) - Offering advice about health and fitness goals or improved road safety to promote loss prevention - Proactive customer outreach and regular customer engagement
Claim Management	Claim settlement Investigation of fraud	- Automated claims management leading to decreasing claim settlement life cycles and increased payout accuracy - Improved fraud detection reduces fraud-related loss positions: anomaly detection, social network analytics and behavioural modelling - Loss reserving aided by AI estimating the value of losses
Asset and Risk Management	Asset allocation Asset liability management Risk control	- Automated investment research with more accurate and detailed market data enables portfolio management to make better-informed decisions due to new insights and more sophisticated analysis of data - Automated risk reporting - Development of robo-advisors for automated asset management - Automated trading systems improve asset allocation

Figure 4: Insurance AI Articles Meeting Relevance Threshold (2000-2021) outlines the number of systematically reviewed articles by year according to the inclusion and exclusion criteria outlined in *Section 3.1*.



The initial screening process included the assessment of 360 articles (following duplicate removal) based on their title, source, and abstract for the presence of the key search terms. In all, 65 articles were included for final review at this stage of the literature search. A backward search of the relevant articles (n = 65) was then conducted, which identified a further 30 articles. The backward search is a popular method of rigorous literature searching within systematic reviews in a range of disciplines including medicine (Mohamadloo *et al.* 2017), law (Siegel *et al.* 2021) and finance (Eckert and Hüsig 2021). The backward search entailed the assessment of the 65 relevant articles' bibliographies for additional articles of relevance to the current review. Based on this rigorous selection process, a total of 95 articles were identified as relevant for the current study (Reference *Appendix A* for the complete database of articles meeting the relevance threshold for inclusion in this systematic review). These results are comprised of ~76% journal articles (n = 72) and ~24% conference papers/proceedings (n = 23). The PRISMA flow diagram depicts the systematic review process (*Figure 5*).

Figure 5: The PRISMA Flow Diagram is a recognised standard for systematic review literature search processes. 'Source' refers to the article inclusion criteria for this systematic review: journal articles and conference papers/proceedings are included.



3.2 Literature Extraction Process

The evaluation of the full-text articles is sub sectioned into three distinct phases in line with the three core contributions of this review. Initially, the articles' applied AI method was distinguished, alongside the prediction task(s) of this AI method. Secondly, the degree of explainability of the AI method employed is analysed. Here, the degree of explainability is evident in the XAI criteria applicable to each AI method employed in each article. Finally, the articles were assessed based on their adherence to the FAIR Data Principles framework (Findability, Accessibility, Interoperability and Reusability). An outline of all AI methods, XAI and FAIR Data criteria is provided in *Table 5.*

The criteria used in evaluating the AI methods' degree of explainability are adapted from Payrovnaziri *et al.* (2020)'s systematic review methodology and modified to suit this review on the insurance industry. The inclusion of the XAI variables and criteria is supported by previous research in XAI, with the criteria synthesised from Mueller *et al.* (2019), Du *et al.* (2019), Carvalho *et al.* (2019) and Payrovnaziri *et al.* (2020). The FAIR Criteria is devised from the guidelines as published by Force11 (2016).

Table 5: AI Methods, XAI Criteria and FAIR Data Criteria used for the systematic analysis of the literature. Data Mining refers to articles which present numerous AI methods, usually comparing predictive powers for specific prediction tasks.

AI Method		**XAI Criteria**	**FAIR Data Criteria**
Bayesian Network	Instance-based	Feature Interaction and Importance	Findable
Clustering	Regression	Attention Mechanism	Accessible
Data Mining*	Reinforcement Learning	(Data) Dimensionality Reduction	Interoperable
Neural Network	Regularisation	Knowledge Distillation & Rule Extraction	Reusable
Decision Tree	Rule-based	Intrinsically Interpretable Models	
Ensemble	Support Vector Machine		
Fuzzy Logic			

3.3 Limitations of the Research

Limitations of the current review are outlined to ensure the validity and reliable reproducibility of results. In particular, the author is unable to access 18 references which Eling *et al.* (2021) presented following their literature search process, while the industry reports reviewed within the same article are not included in the current systematic review. The lack of industry reports' analysis in this paper leads to an absence of articles concerning the Support Activities stage on the IVC. In Eling *et al.* (2021)'s research, all articles found pertaining to insurance companies' Support Activities were industry reports.

Industry reports were not included in this paper as access to articles with complete methodological processes outlined is pertinent to the current systematic review, a section which industry reports regularly omit in their publications. The inclusion of academic articles and conference articles ensures the methods of AI integration in each of the reviewed articles is outlined, in particular a coherent methodology discussion which can be assessed using the criteria outlined in this paper.

As alluded to above, alternative databases to those inaccessible to the author were chosen. The following alternative databases include similar published literature to that which could not be readily accessed: 1) IEEE Xplore was searched as an alternative to ABI/INFORM Collection and, 2) ACM Digital Library was searched as an alternative to EBSCOhost's Computer Science database. These substitute decisions were developed in line with findings from Gusenbauer and Haddaway (2020) which compared the validity of different research algorithms, with a specific focus on their use in systematic review research. Each of the databases used in this paper satisfy desired criteria for use in systematic review research retrieval.

4. Systematic Review Results

4.1 AI Methods and Prediction Tasks

The systematically chosen articles are first assessed based on the AI method employed and associated prediction task, with a focus on then distinguishing the degree of explainability evident in the literature. The stage of the IVC each article refers to is also clarified in the systematic research findings. Research on AI's use along the IVC over the twenty-one-year period of this review revealed AI is popular at every stage of the IVC, with the exception of insurance companies' Support Activities. Such activities include general HR, IT and Public Relations departments insurance companies. A viable reason for the lack of articles concerned with this stage of the IVC is that Eling *et al.* (2021)'s study found articles on this subject through their review of industry reports, which the present systematic review did not include in the systematic review. The Underwriting and Pricing stage reveals significant research results (~ 43%), with Claim Management (~ 31%) also making extensive use of AI methods for fraud management and identification in particular.

Table 6 lists all the articles alongside the AI method employed and prediction task. A range of AI methods are used in the articles including 1) Ensemble, 2) Neural Network (NN), 3) Clustering, 4) Regression (Linear and Logistic), 5) Fuzzy Logic, 6) Bayesian Network (BN), 7) Decision Tree, 8) Support Vector Machine (SVM). Other methods used include Instance- and Rule-based, Regularisation and Reinforcement Learning. AI methods classified as 'Data Mining' signify that the article utilised numerous AI methods, usually comparing different AI methods' use for the prediction task. The most popular AI method used is Ensemble (~ 23%), with both NNs (~ 18%) and Clustering (~ 14%) also proving popular.

The line of insurance business the research in each article refers to is also classified, with non-life insurance lines returning a high number of articles in the systematic review (~ 55%). Motor insurance prediction problems are popular areas of research, including driving behaviour classification and automobile insurance fraud (~ 45%). Articles concerning insurers' life-business shows health(care) insurance as a popular area of research (~ 13%), with health insurance fraud prevention and the classification of health insureds the most prominent research areas.

Table 6: AI Methods and Prediction Tasks. Abbreviations in *Table 6* are outlined in the Abbreviations section of this paper.

		AI Method	Prediction Task(s)	Life/Non-Life	Line of Insurance Business
Marketing					
1	Desik *et al.* (2016)	Regression	Develop a predictive modelling solution to aid the identification of the best insurance product group for current insurance product group of customers	-	-
2	Fang *et al.* (2016)	Ensemble	Prediction of insurance customer profitability	Life	Health
3	Larivière and Van den Poel (2005)	Ensemble	Prediction of customer retention and profitability	-	-
4	Lin *et al.* (2017)	Ensemble	Classification to enhance marketing of insurance products	Life	-
5	Morik *et al.* (2002)	Data Mining	Extraction of low-level knowledge data to answer high-level questions on customer acquisition, customer up and cross selling and customer retention within insurance companies	-	-
Product Development					
6	Alshamsi (2014)	Ensemble	Prediction of automobile insurance policies chosen by customers using Random Forest (RF)	Non-life	Motor
7	Karamizadeh and Zolfagharifar (2016)	Clustering	*K*-means used to identify clusters which contribute to the profit & loss of auto insurance companies	Non-life	Motor
8	Khodairy and Abosamra (2021)	Neural Network	Driving behaviour classification	Non-life	Motor
9	Shah and Guez (2009)	Neural Network	Calculation of life expectancy (mortality forecasting) based on the individual's health status	Life	Health
10	Sheehan *et al.* (2017)	Bayesian Network	BN risk estimation approach for the emergence of new risk structures, including autonomous vehicles	Non-life	Motor and Product Liability
Sales and Distribution					
11	Desik and Behera (2012)	Decision Tree	Creation of business rules from customer-led data to improve insurer competitiveness	-	-
12	Jeong *et al.* (2018)	Data Mining	Association between policyholder switching after a claim and the associated change in premium	Non-life	Motor
13	Tillmanns *et al.* (2017)	Bayesian Network	Selection of promising prospective insurance customers from a vendor's address list	-	-
14	Wang (2020)	Ensemble	Prediction of auto renewal using RF	Non-life	Motor
15	Yang *et al.* (2006)	Ensemble	Ensemble of DTs used to maximise the expected net profit of customers	-	-
16	Zahi and Achchab (2019)	Clustering	Grouping of health insured population	Life	Health
17	Zhang and Kong (2020)	Bayesian Network	Estimation of insurance product recommendation	-	-
Underwriting and Pricing					
18	Aggour *et al.* (2006)	Fuzzy Logic	Encoded the underwriting guidelines to automate the underwriting procedures of long-term care and life-insurance policies	Life	Long Term Care
19	Baecke and Bocca (2017)	Regression	Assess the enhanced accuracy of risk selection predictive models utilising driving behaviour variables in addition to traditional accident risk predictors	Non-life	Motor
20	Bian *et al.* (2018)	Ensemble	Ensemble learning-based approach to obtain information on a user's risk classification which informs the compensation payout	Non-life	Motor
21	Biddle *et al.* (2018)	Data Mining	Prediction of the applications of exclusions in life insurance policies when automated underwriting methods are employed	Life	-
22	Bonissone *et al.* (2002)	Fuzzy Logic	Automation of underwriting practices	-	-
23	Boodhun and Jayabalan (2018)	Neural Network	Predict the risk level of life insurance applicants	Life	-

		AI Method	Prediction Task(s)	Life/Non-Life	Line of Insurance Business
24	Carfora *et al.* (2019)	Clustering	Evaluation of UBI automobile insurance policies	Non-life	Motor
25	Cheng *et al.* (2011)	Support Vector Machine	Evaluation of loss risk & development of criteria for optimal insurance deductible decision making	Non-life	Construction
26	Christmann (2004)	Ensemble	Indirect estimation of the pure premium in motor vehicle insurance	Non-Life	Motor
27	David (2015)	Regression	Use of the GLM to establish policyholders' pure premium	Non-life	Motor
28	Denuit and Lang (2004)	Regression	GAMs used for rate-making	Non-life	Motor
29	Deprez *et al.* (2017)	Ensemble	Mortality modelling using boosting regression techniques	Life	-
30	Devriendt *et al.* (2021)	Regularisation	LASSO penalty development to aid regularisation techniques in ML	-	-
31	Gan (2013)	Clustering	Selection of representative policies for the assessment of variable annuity policy pricing	Life	-
32	Gan and Huang (2017)	Clustering	Valuation of variable annuity policies	Life	-
33	Gan and Valdez (2017)	Reinforcement Learning	Monte Carlo-based modelling for variable annuity portfolios	Life	-
34	Guelman (2012)	Ensemble	Gradient Boosting Trees used to predict insurance losses	Non-life	Motor
35	Gweon *et al.* (2020)	Ensemble	Bias-corrected bagging method used to improve predictive performance of regression trees	Non-life	-
36	(Huang and Meng 2019)	Regression	Risk probability prediction based on telematics driving data	Non-life	Motor
37	Jain *et al.* (2019)	Ensemble	Risk assessment of potential policyholders using risk scores within numerous ensembles of AI methods	Life	-
38	Jiang *et al.* (2018)	Data Mining	Novel model for analysis of imbalanced datasets in end-to-end insurance processes	Life	-
39	Joram *et al.* (2017)	Rule-based	Knowledge-based system to enhance life underwriting processes	Life	-
40	Kašćelan *et al.* (2016)	Clustering	Assessment and classification of premiums	Non-life	Motor
41	Kieu *et al.* (2018)	Clustering	Deal with inadequately labelled data trajectories with drivers' identifiers	Non-life	Motor
42	Kumar *et al.* (2010)	Support Vector Machine	Prediction of claims which need reworking due to errors	Life	Health
43	Kwak *et al.* (2020)	Ensemble	Driver identification using RF	Non-life	Motor
44	Lin (2009)	Neural Network	Price the correct premium rate for 'in-between' risks between predefined tariff rates	Non-life	Property & Casualty
45	Liu *et al.* (2014)	Ensemble	Adaboost to predict claim frequency of auto insurance	Non-life	Motor
46	Neumann *et al.* (2019)	Decision Tree	Prediction of insurance customers' decisions following an automobile accident	Non-life	Motor
47	Poon (2019)	Neural Network	Prediction of loss development and segmentation of risk pricing	Non-life	Property & Casualty
48	Sakthivel and Rajitha (2017)	Neural Network	Prediction of an insurance portfolio's claim frequency for forthcoming years	Non-life	Motor
49	Samonte *et al.* (2018)	Neural Network	Automatic multi-class labelling of ICD-9 codes of patient notes	Life	Health
50	Schelldorfer and Wuthrich (2019)	Ensemble	Embedding of GLM into NN to enhance treatment of categorical variables within actuarial models	Non-life	Motor
51	Sevim *et al.* (2016)	Neural Network	Determination of litigation risks for accounting professional liability insurance	Non-life	Professional Liability
52	Siami *et al.* (2020)	Instance-Based	Unsupervised pattern recognition framework for mobile telematics data to propose a solution to unlabelled telematics data	Non-life	Motor

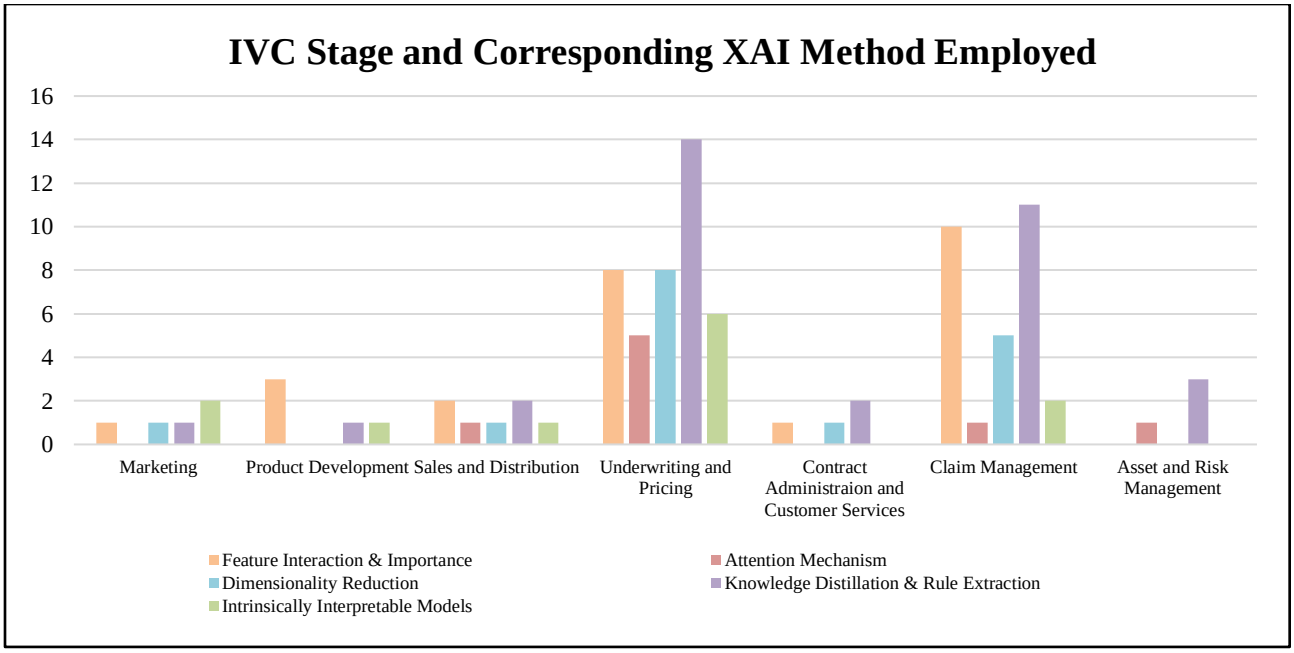
		AI Method	Prediction Task(s)	Life/Non-Life	Line of Insurance Business
53	Smith *et al.* (2000)	Neural Network	NNs used to classify policy holders as likely to renew or terminate, to aid the achievement of maximum potential profitability for the insurance company	Non-life	Motor
54	Wei and Dan (2019)	Support Vector Machine	Stock price prediction	Non-life	Agriculture
55	Yan and Bonissone (2006)	Neural Network	Classification to enhance NN functionality for automated insurance underwriting	-	-
56	Yan *et al.* (2020)	Data Mining	Rating model for UBI automobile insurance rates	-	-
57	Yang *et al.* (2018)	Ensemble	Gradient Boosting Trees used to predict insurance premiums	Non-life	Motor
58	Yeo *et al.* (2002)	Clustering	Optimisation of insurance premium pricing	Non-life	Motor
Contract Administration and Customer Services					
59	Ravi *et al.* (2017)	Fuzzy Logic	Creation of association rules which analyses customer grievances and summarises them	-	-
60	Sadreddini *et al.* (2021)	Clustering	Prediction of airline customer clusters and appropriate Cancellation Protection Service insurance fee per customer group	Non-life	Airline
61	Vassiljeva *et al.* (2017)	Neural Network	Automobile insurance customers' risk estimate using ANN to inform contract development	Non-life	Motor
62	Vaziri and Beheshtinia (2016)	Fuzzy Logic	Value creation for insurance customers	Life	-
Claim Management					
63	Baudry and Robert (2019)	Ensemble	Estimation of outstanding liabilities on a given policy using an ensemble of regression trees	-	-
64	Belhadji *et al.* (2000)	Regression	Calculate the probability of fraud in insurance files	Non-life	Motor
65	Benedek and László (2019)	Data Mining	Identification of fraud indicators	Non-Life	Motor
66	Bermúdez *et al.* (2008)	Bayesian Network	Bayesian skewed logit model used to fit an insurance database (binary data)	Non-life	Motor
67	Cao and Zhang (2019)	Instance-Based	SOFM NN used to extract characteristics of medical insurance fraud behaviour	Life	Health
68	Dhieb *et al.* (2019)	Ensemble	XGBoost used to detect automobile insurance fraudulent claims	Non-life	Motor
69	Frees and Valdez (2008)	Regression	Assessment of claim frequency	Non-life	Motor
70	Ghani and Kumar (2011)	Support Vector Machine	Error detection in insurance claims	Life	Health
71	Ghorbani and Farzai (2018)	Clustering	Detection of fraud patterns	Non-life	Motor
72	Herland *et al.* (2018)	Ensemble	Medicare provider claims fraud	Life	Health
73	Johnson and Khoshgoftaar (2019)	Neural Network	Automation of fraud detection using ANN	Life	Health
74	Kose *et al.* (2015)	Clustering	Detection of fraudulent claims	Life	Health
75	Kowshalya and Nandhini (2018)	Data Mining	Fraudulent claim detection	Non-life	Motor
76	Kyu and Woraratpanya (2020)	Neural Network	CNN used to prevent claims leakage	Non-life	Motor
77	Lau and Tripathi (2011)	Data Mining	Association Rules' provision of actionable business insights for insurance claims data	Non-life	Liability

		AI Method	Prediction Task(s)	Life/Non-Life	Line of Insurance Business
78	Lee *et al.* (2020)	Regression	GLM and GAM used in NLP to extract variables from text and use these variables in claims analyses	Non-life	Property & Casualty
79	Li *et al.* (2018)	Ensemble	Random Forest for automobile insurance fraud detection	Non-life	Motor
80	Matloob *et al.* (2020)	Data Mining	Fraud detection	Life	Health
81	Pathak *et al.* (2005)	Fuzzy Logic	To distinguish whether fraudulent actions are involved in insurance claims settlement	-	-
82	Smyth and Jørgensen (2002)	Regression	GLM to model insurance costs' dispersion	Non-life	Motor
83	Sun *et al.* (2018)	Data Mining	Determination of joint medical fraud through reducing the occurrence of false positives caused by non-fraudulent abnormal behaviour	Life	Health
84	Supraja and Saritha (2017)	Fuzzy Logic	Utilising fuzzy rule-based techniques to improve fraud detection	Non-life	Motor
85	Tao *et al.* (2012)	Fuzzy Logic	DFSVM used to solve the issue of misdiagnosed fraud detection due to the 'overlap' problem in insurance fraud samples	Non-life	Motor
86	Verma *et al.* (2017)	Clustering	*k*-means used to increase performance and reduce complexity of the model	Life	Health
87	Viaene *et al.* (2002)	Regression	Fraud detection	Non-life	Motor
88	Viaene *et al.* (2004)	Ensemble	Adaboost used in insurance claim fraud detection	Non-life	Motor
89	Viaene *et al.* (2005)	Bayesian Network	NN for fraud detection	Non-life	Motor
90	Wang and Xu (2018)	Neural Network	NN used to detect automobile insurance fraud	Non-life	Motor
91	Xu *et al.* (2011)	Ensemble	Random rough subspace method	Non-life	Motor
Asset and Risk Management					
92	Cheng *et al.* (2020)	Neural Network	Optimal reinsurance and dividend strategies for insurance companies	-	-
93	Ibiwoye *et al.* (2012)	Neural Network	Insurer insolvency prediction	-	-
94	Jin *et al.* (2021)	Neural Network	Determine the optimal insurance, reinsurance, and investment strategies of an insurance company	-	-
95	Kiermayer and Weiß (2021)	Clustering	Grouping of insurance contracts	Life	Life

4.2 XAI Categories along the IVC

The following categories of XAI methods are highlighted within the article database; 1) Feature Interaction and Importance, 2) Attention Mechanism, 3) Dimensionality Reduction, 4) Knowledge Distillation and Rule Extraction, and 5) Intrinsically Interpretable Models. *Figure 6* shows each stage on the IVC and the corresponding XAI method employed in the reviewed articles. The XAI methods' interpretability techniques are then categorised into 1) intrinsic vs. post-hoc, 2) local vs. global and 3) model-specific vs. model-agnostic. According to the reviewed articles, most of the research on AI applications in insurance is concerned with Knowledge Discovery and Distillation, which is also grouped with Rule Extraction (~ 36%) XAI methods for the purposes of the current review.

Figure 6: IVC Stage and Corresponding XAI Method Employed presents the seven IVC stages assessed in the systematically chosen articles and the XAI method used in their methodology. Support Activities is not included in this paper as no articles presented in the systematic literature search which presented prediction tasks in line with insurance companies' Support Activities.



Feature Interaction and Importance

Analysing AI models' input features importance and interaction is a popular XAI method, with ~ 27% of reviewed articles utilising this method. The determination of features' importance contributed to the development of thorough AI methods to complete many prediction tasks at each stage on the IVC. Smith *et al.* (2000) utilise Artificial Neural Networks (ANN) to gain an insight into customer policies which were likely to renew or terminate at the close of the policy period through analysing those factors which contribute to policy termination. This assessment of optimal premium pricing through data mining and ML methods instructs research on insurance customer retention and profitability. Also addressing customer retention is Larivière and Van den Poel (2005)'s research which explored three predictor variables which encompass potential explanatory variables to inform insurance customer retention. Their RF model provides an importance measure between the explanatory and dependence variables for the prediction task.

Claim management and insurance fraud detection is an area which benefits from analysing the interaction and importance of feature inputs in AI applications through the isolation of important features which contribute to fraud (Belhadji *et al.* 2000). Similarly, Tao *et al.* (2012) avoid the curse of dimensionality through using the kernel function for SVMs in their XAI approach for insurance fraud identification, while Supraja and Saritha (2017) use this XAI method to ready their data for automobile fraud detection using fuzzy rule-based predictive techniques.

Feature interaction and importance is also useful in assessing risk across a wide range of insurance activities and informing underwriting and pricing of premiums. Biddle *et al.* (2018) add to literature on automated underwriting in life insurance applications using the XAI method of Feature Interaction and Importance. Recursive Feature Elimination is used to reduce the feature space through iteratively wrapping and training a classifier on several feature subsets and then provides feature rankings for each subset. Premium pricing of automobile insurance is researched in Yeo *et al.* (2002)'s research, where cluster grouping of policyholders according to relative features aids in determining the price sensitivity of policyholder groups to premium prices.

Table 7: XAI Methods and their approach in the articles is outlined, with the additional XAI assessment of (i) intrinsic or post-hoc, (ii) local or global, and (iii) model-specific or model-agnostic interpretability methods. Abbreviations in *Table 7* are outlined in the Abbreviations section of this paper.

		XAI Category	XAI Approach	Intrinsic/Post-hoc	Local/Global	Model-specific/agnostic
Marketing						
1	Desik *et al.* (2016)	Dimensionality Reduction	Identification of relevant data clusters to inform model development for differing product groups	Post-hoc	Local	Model-agnostic
2	Fang *et al.* (2016)	Intrinsically Interpretable Model	RF regression	Intrinsic	Global	Model-specific
3	Lariviere and van der Poel (2005)	Feature Interaction and Importance	Exploration of three major predictor categories as explanatory variables	Intrinsic	Local	Model-specific
4	Lin *et al.* (2017)	Intrinsically Interpretable Model	RF providing automatic feature selection which aids interpretability of the model	Intrinsic	Global	Model-specific
5	Morik *et al.* (2002)	Knowledge Distillation and Rule Extraction	Bridge the gap between databases and their users through implementing KDD methods	Intrinsic	Local	Model-specific
Product Development						
6	Alshamsi (2014)	Feature Interaction and Importance	Classification of data into different sets according to different policy options available	Intrinsic	Local	Model-specific
7	Karamizadeh and Zolfagharifar (2016)	Intrinsically Interpretable Model	Pattern recognition with clustering algorithms to find missing data with the goal of minimising insurance losses	Intrinsic	Global	Model-specific
8	Khodairy and Abosamra (2021)	Feature Interaction and Importance	Extraction of relevant features	Post-hoc	Local	Model-agnostic
9	Shah and Guez (2009)	Feature Interaction and Importance	NN proposed as better predictor of life expectancy than Lee-Carter model due to ability to adapt for each sex and each cause of life expectancy through a learning algorithm using historical data	Post-hoc	Local	Model-agnostic
10	Sheehan *et al.* (2017)	Knowledge Distillation and Rule Extraction	Determination of causal and probabilistic dependencies through subjective assumptions (of the data)	Intrinsic	Local	Model-specific
Sales and Distribution						
11	Desik and Behera (2012)	Feature Interaction and Importance	CHAID used to create groups and gain understanding of their impact on the dependent variable	Intrinsic	Local	Model-specific
12	Jeong *et al.* (2018)	Knowledge Distillation and Rule Extraction	Association rule learning to identify relationships among variables	Intrinsic	Global	Model-specific
13	Tillmanns et al. (2017)	Feature Interaction and Importance	PCA used to reduce the features dimensionality and reduces the chance of overfitting	Post-hoc	Local	Model-agnostic
14	Wang (2020)	Dimensionality Reduction	Removal of dataset features which have no bearing on the customer's likelihood to renew	Intrinsic	Local	Model-specific
15	Yang *et al.* (2006)	Knowledge Distillation and Rule Extraction	Development of postprocessing step to extract actionable knowledge from DTs to obtain action which are associated with attribute-value changes	Intrinsic	Local	Model-specific
16	Zahi and Achchab (2019)	Intrinsically Interpretable Model	Clustering the insured population using *k*-means	Intrinsic	Global	Model-specific
17	Zhang and Kong (2020)	Attention Mechanism	Parameter optimisation for NB model	Post-hoc	Local	Model-agnostic
Underwriting and Pricing						
18	Aggour et al. (2006)	Feature Interaction and Importance	Use of NLP and explanation of the interaction of different model features which alters the model	Intrinsic	Global	Model-specific
19	Baecke and Bocca (2017)	Feature Interaction and Importance	Stepwise feature selection	Intrinsic	Global	Model-specific
20	Bian *et al.* (2018)	Dimensionality Reduction	Found the 5 most relevant features to inform driving behaviour	Intrinsic	Local	Model-specific
21	Biddle *et al.* (2018)	Feature Interaction and Importance	Recursive Feature Elimination to provide feature rankings for feature subsets	Post-hoc	Global	Model-agnostic
22	Bonssione *et al.* (2002)	Knowledge Distillation and Rule Extraction	Fuzzy rule-based decision systems used to encode risk classification of complex underwriting tasks	Intrinsic	Local	Model-specific

		XAI Category	XAI Approach	Intrinsic/Post-hoc	Local/Global	Model-specific/agnostic
23	Boodhun and Jayabalan (2018)	Dimensionality Reduction	Correlation-Based Feature Selection and PCA	Intrinsic	Local	Model-specific
24	Carfora *et al.* (2019)	Intrinsically Interpretable Model	Identification of driver behaviour using ML algorithms	Intrinsic	Global	Model-specific
25	Cheng *et al.* (2011)	Knowledge Distillation and Rule Extraction	Development of loss prediction model using the ESIM	Intrinsic	Global	Model-specific
26	Christmann (2004)	Dimensionality Reduction	Exploitation of knowledge from certain characteristics of datasets to estimate conditional probabilities and conditional expectations given the knowledge of the variable representing the pure premium	Intrinsic	Local	Model-specific
27	David (2015)	Dimensionality Reduction	Use of policyholders' relevant characteristics to determine the pure premium	Intrinsic	Local	Model-specific
28	Denuit and Lang (2004)	Knowledge Distillation and Rule Extraction	Bayesian GAMs developed using MCAM inference	Intrinsic	Local	Model-specific
29	Deprez *et al.* (2017)	Attention Mechanism	Back-testing parametric mortality models	Post-hoc	Global	Model-agnostic
30	Devriendt *et al.* (2021)	Knowledge Distillation and Rule Extraction	Development of SMuRF algorithm to allow for Sparse Multi-type Regularised Feature modelling	Intrinsic	Global	Model-specific
31	Gan (2013)	Knowledge Distillation and Rule Extraction	Gaussian Process Regression employed to value variable annuity policies	Intrinsic	Local	Model-specific
32	Gan and Huang (2017)	Knowledge Distillation and Rule Extraction	Kriging Regression method employed to value variable annuity policies	Intrinsic	Local	Model-specific
33	Gan and Valdez (2018)	Knowledge Distillation and Rule Extraction	Generalised Beta of the Second Kind (GB2) Regression method employed to value variable annuity policies	Intrinsic	Local	Model-specific
34	Guelman (2012)	Intrinsically Interpretable Model	Interpretable results given by the simple linear model through showcasing the relative influence of the input variables and their partial dependence plots	Intrinsic	Global	Model-specific
35	Gweon *et al.* (2020)	Knowledge Distillation and Rule Extraction	Bagging creates several regression trees which fits a bootstrap sample of the training data and makes a prediction through averaging the predicted outcomes from the bootstrapped trees	Post-hoc	Local	Model-agnostic
36	Huang and Meng (2019)	Dimensionality Reduction	Variables are binned to discretise continuous variables and construct tariff classes with significant predictive effects to improve interpretability of UBI predictive models	Post-hoc	Intrinsic	Model-agnostic
37	Jain *et al.* (2019)	Feature Interaction and Importance	Using WEKA software, the dimensional feature set was reduced for use	Intrinsic	Global	Model-specific
38	Jiang *et al.* (2018)	Feature Interaction and Importance	Imbalanced data trend forecasting using learning descriptions and sequences and adjusting the CPLF	Post-hoc	Local	Model-specific
Underwriting and Pricing						
39	Joram *et al.* (2017)	Knowledge Distillation and Rule Extraction	Containment of the sets of rules with similar purpose and/or structure which defines the knowledge bases	Intrinsic	Global	Model-agnostic
40	Kašćelan *et al.* (2016)	Intrinsically Interpretable Model	Clustering provides homogeneity within classifications of risk and heterogeneity between risk classifications	Intrinsic	Global	Model-specific
41	Kieu *et al.* (2018)	Intrinsically Interpretable Model	Gradient Boosting DTs used to classify (unlabelled) trajectories	Post-hoc	Local	Model-specific
42	Kumar *et al.* (2010)	Dimensionality Reduction	Frequency-based feature selection technique	Intrinsic	Global	Model-specific
43	Kwak *et al.* (2020)	Dimensionality Reduction	Reduction of feature values' noise (normalisation of sensing data)	Intrinsic	Local	Model-specific
44	Lin (2009)	Attention Mechanism	Use of premium rate determination rules as network inputs in the BPNN to create the 'missing rates' of in-between risks	Post-hoc	Local	Model-specific
45	Liu *et al.* (2014)	Dimensionality Reduction	Reduction of claim frequency prediction problem to multi-class problem	Post-hoc	Global	Model-specific
46	Neumann *et al.* (2019)	Intrinsically Interpretable Model	DT provides interpretable and transparent flow of insurance customers' potential decision pathways	Intrinsic	Local	Model-specific
47	Poon (2019)	Knowledge Distillation and Rule Extraction	Combination of simple linear weights and residual components to replicate non-linear effects to resemble a fully parametrised PPCI-like (Payments per Claim Incurred) model	Intrinsic	Local	Model-specific
48	Sakthivel and Rajitha (2017)	Knowledge Distillation and Rule Extraction	Built a predictive model using previous Bayesian credibility inputs to predict the value of another field	Post-hoc	Local	Model-specific
49	Samonte *et al.* (2018)	Attention Mechanism	NLP used for document classification of medical record notes, with RNNs employed to encode vectors in Bi-LTSM model	Intrinsic	Local	Model-specific

		XAI Category	XAI Approach	Intrinsic/Post-hoc	Local/Global	Model-specific/agnostic
50	Schelldorfer and Wuthrich (2019)	Knowledge Distillation and Rule Extraction	Model compression to aid actuarial models' treatment of categorical variables	Post-hoc	Global	Model-agnostic
51	Sevim *et al.* (2016)	Attention Mechanism	Model is developed from the relationships between the variables gained from previous data and then tested	Post-hoc	Local	Model-specific
52	Siami *et al.* (2020)	Feature Interaction and Importance	SOM to reduce data complexity	Intrinsic	Global	Model-specific
53	Smith *et al.* (2000)	Feature Interaction and Importance	Assessed the variables of relevance to the current task through rejecting variables with $x2 < 3.92$	Post-hoc	Local	Model-agnostic
54	Wei and Dan (2019)	Attention Mechanism	Parameter optimisation for SVM model	Intrinsic	Global	Model-specific
55	Yan and Bonissone (2006)	Knowledge Distillation and Rule Extraction	Comparison of four NN models for automated insurance underwriting	Post-hoc	Local	Model-specific
56	Yan *et al.* (2020)	Knowledge Distillation and Rule Extraction	Combination of the CNN and HVSVM models to create a model with higher discrimination accuracy that either model presents alone	Post-hoc	Global	Model-specific
57	Yang *et al.* (2018)	Intrinsically Interpretable Model	TDBoost package provides interpretable results	Intrinsic	Local	Model-specific
58	Yeo *et al.* (2002)	Feature Interaction and Importance	Grouping of important clusters to input in NN model for insurance retention rates and price sensitivity prediction	Intrinsic	Local	Model-specific
Contract Administration and Customer Services						
59	Ravi *et al.* (2017)	Knowledge Distillation and Rule Extraction	Treatment of each variable as having a certain degree of membership with certain rules to categorise complaints	Intrinsic	Global	Model-specific
60	Sadreddini *et al.* (2021)	Feature Interaction and Importance	Cancellation Protection Service insurance fee is calculated based on the relevant weight of each cluster	Intrinsic	Global	Model-specific
61	Vassiljeva *et al.* (2017)	Dimensionality Reduction	Only relevant parameters are considered in the ANN model	Intrinsic	Local	Model-specific
62	Vaziri and Baheshtina (2016)	Knowledge Distillation and Rule Extraction	Development of integrated ML model to carry out the prediction task	Intrinsic	Local	Model-specific
Claim Management						
63	Baudry and Robert (2019)	Feature Interaction and Importance	Definition of policy subsets within the synthetic dataset	Post-hoc	Local	Model-agnostic
64	Belhadji *et al.* (2000)	Feature Interaction and Importance	Regression used to isolate significant contributory variables in fraud	Intrinsic	Local	Model-specific
65	Benedek and Laszlo (2019)	Intrinsically Interpretable Model	Comparison of various intrinsic AI methods for fraud indicator identification	Intrinsic	Local	Model-specific
66	Bermudez *et al.* (2008)	Knowledge Distillation and Rule Extraction	Use of a skewed logit model to more accurately classify fraudulent insurance claims	Post-hoc	Global	Model-agnostic
67	Cao and Zhang (2019)	Dimensionality Reduction	PCA in the reduction of data's dimensionality	Post-hoc	Local	Model-agnostic
68	Dhieb *et al.* (2019)	Dimensionality Reduction	Extraction of relevant features	Post-hoc	Global	Model-specific
69	Frees and Valdez (2008)	Knowledge Distillation and Rule Extraction	Comparison of various fitted models which summarise all the covariates' effects on claim frequency	Intrinsic	Global	Model-specific
70	Ghani and Kumar (2011)	Knowledge Distillation and Rule Extraction	Development of an interactive prioritisation component to aid auditors in their fraud detection	Post-hoc	Local	Model-specific
71	Ghorbani and Farzai (2018)	Knowledge Distillation and Rule Extraction	Definition of rules based on each cluster to determine future fraud propensity (using WEKA)	Intrinsic	Global	Model-specific
72	Herland *et al.* (2018)	Feature Interaction and Importance	Removed unnecessary data features	Intrinsic	Local	Model-specific
73	Johnson (2019)	Feature Interaction and Importance	Class imbalance within the dataset is rectified using one-hot encoding	Post-hoc	Local	Model-specific
74	Kose *et al.* (2015)	Knowledge Distillation and Rule Extraction	Development of an electronic fraud & abuse detection model	Post-hoc	Global	Model-agnostic
75	Kowshalya and Nandhini (2018)	Feature Interaction and Importance	Classifier construction using NB	Intrinsic	Local	Model-specific

		XAI Category	XAI Approach	Intrinsic/Post-hoc	Local/Global	Model-specific/agnostic
76	Kyu and Woraratpanya (2020)	Feature Interaction and Importance	Fine-tuning of the dataset	Post-hoc	Local	Model-specific
77	Lau and Tripathi (2011)	Knowledge Distillation and Rule Extraction	Development of Association Rules function for Workers' Compensation claim data analysis	Intrinsic	Global	Model-specific
78	Lee *et al.* (2020)	Knowledge Distillation and Rule Extraction	Transformation of words to vectors, where each vector represents some feature of the word	Intrinsic	Local	Model-specific
79	Li *et al.* (2018)	Dimensionality Reduction	PCA used to transform data at each node to another space when computing the best split at that node	Intrinsic	Global	Model-specific
80	Matloob *et al.* (2020)	Knowledge Distillation and Rule Extraction	Sequence generation to inform predictive model for fraudulent behaviour	Intrinsic	Local	Model-specific
81	Pathak *et al.* (2005)	Knowledge Distillation and Rule Extraction	Mimic the expertise of the human insurance auditors in real life insurance claim settlement scenarios	Post-hoc	Local	Model-agnostic
82	Smyth and Jorgensen (2002)	Intrinsically Interpretable Model	Modelling of insurance costs' dispersion and mean	Intrinsic	Local	Model-specific
83	Sun *et al.* (2018)	Feature Interaction and Importance	Formulation of compact clusters of individual behaviour in a large dataset	Intrinsic	Local	Model-specific
84	Supraja and Saritha (2017)	Feature Interaction and Importance	*k*-means clustering used to prepare dataset prior to FL technique application	Intrinsic	Local	Model-specific
85	Tao *et al.* (2012)	Feature Interaction and Importance	Avoidance of curse of dimensionality problem through kernel function use for SVM's calculation	Post-hoc	Global	Model-agnostic
86	Verma *et al.* (2017)	Knowledge Distillation and Rule Extraction	Association rule learning to identify frequent fraud occurring patterns for varying groups	Intrinsic	Local	Model-specific
87	Viaene *et al.* (2002)	Dimensionality Reduction	Removal of fraud indicators with 10 or less instances to aid model convergence and stability during estimation	Intrinsic	Global	Model-specific
88	Viaene *et al.* (2004)	Attention Mechanism	Computation of the relative importance (weight) of individual components of suspicious claim occurrences	Intrinsic	Global	Model-specific
89	Viaene *et al.* (2005)	Feature Interaction and Importance	Determination of relevant inputs for the NN model	Post-hoc	Local	Model-agnostic
90	Wang and Xu (2018)	Dimensionality Reduction	Extraction of text features hiding in the text descriptions of claims (Latent Dirichlet Allocation-based deep learning for text analytics)	Post-hoc	Local	Model-agnostic
91	Xu *et al.* (2011)	Knowledge Distillation and Rule Extraction	Random rough subspace method incorporated into NN to detect insurance fraud	Intrinsic	Global	Model-specific
Asset and Risk Management						
92	Cheng *et al.* (2020)	Knowledge Distillation and Rule Extraction	Development of deep learning Markov chain approximation method (MCAM)	Intrinsic	Global	Model-specific
93	Ibiwoye *et al.* (2012)	Attention Mechanism	Tuning of the NN	Intrinsic	Local	Model-specific
94	Jin *et al.* (2021)	Knowledge Distillation and Rule Extraction	MCAM to estimate the initial guess of the NN	Intrinsic	Global	Model-specific
95	Kiermayer and Weiß (2021)	Knowledge Distillation and Rule Extraction	Approximation of representative portfolio groups to then nest in NN	Post-hoc	Local	Model-specific

Attention Mechanism

The Attention Mechanism within an AI model primarily attempts to find a set of positions in a sequence with the most relevant information on a prediction task (Payrovnaziri *et al.* 2020), which in turn enhances interpretability, according to Mascharka *et al.* (2018).

In line with the current review, Attention Mechanism is used to compute the weight of claim occurrences to inform fraud detection (Viaene *et al.* 2004) and inform insurer insolvency prediction (Ibiwoye *et al.* 2012). Lin (2009) applies Attention Mechanism in their determination of premium rates of 'in-between' risks through weight classification of different tariff classes. The method also aids in the determination of litigation risk of liability insurance within the accountancy profession, as Sevim *et al.* (2016) incorporate Attention Mechanism in their development of an ANN model, while Deprez *et al.* (2017) apply Attention Mechanism to mortality modelling through back-testing parametric mortality models. Samonte *et al.* (2018) use this XAI method for automatic document classification of medical record notes using NLP. The enhancement of the Hierarchical Attention Network model (EnHAN) assigns topics for each word in a given text and learns topical word embedding in a hierarchical manner. Topical word embedding models solve the multi-label, multi-class classification problem within medical records to inform cluster processes for billing and insurance claims.

Wei and Dan (2019) apply Attention Mechanism to parameter optimisation of SVM features, while Zhang and Kong (2020) also optimised parameters for input in NB model to inform insurance product recommendations. In terms of sequence generation, this XAI method was used by Matloob *et al.* (2020) to inform their predictive model for fraudulent behaviour in health insurance.

Dimensionality Reduction

Researchers typically use dimensionality reduction techniques in order to reduce the set of features inputted in the model principally to improve a model's efficiency. Kumar *et al.* (2010), for instance, use a frequency-based feature selection technique to reduce the dataset dimensions. This action aided in developing a model for error prevention in health insurance claims processing through reducing data storage requirements and improved model execution time. They found that using a lower frequency threshold and limiting the input feature improved the predictive accuracy. Finding similar results in terms of improved predictive accuracy, Li *et al.* (2018) use Principal Component Analysis (PCA) to increase the diversity of each of the 100 trees used in a RF model. This action improves the overall accuracy of the algorithm. In this instance, PCA transforms the data at each node to another space when computing the best split at that node which contributed to satisfactory feature selection in the development of the RF algorithm for fraud detection. PCA is also used in Underwriting and Pricing of life insurance through model development for risk assessment of life insurance customers (Boodhun and Jayabalan 2018).

For the popular prediction tasks related to automobile insurance, the reduction of dataset dimensionality is also useful. Liu *et al.* (2014) reduce their large claim frequency prediction to a multi-class prediction problem to aid the eventual implementation of Adaptive Boosting (AdaBoost) to automobile insurance data. The act of reducing the number of frequency classes contributes to AdaBoost presenting as superior to SVM, NN, DTs and GLM in terms of prediction ability and interpretability. Huang and Meng (2019) bin variables to approximate continuous variables in the dataset and construct tariff classes with high-level predictive power which enhances the model's accuracy and predictive power in the classification of usage-based insurance (UBI) products. An ANN model is optimised in Vassiljeva *et al.* (2017) to inform automobile contract development through assessing drivers' risk, while Bian *et al.* (2018) reduced their data dimensions to include only the five most relevant factors in determining drivers' behaviour.

Other stages on the IVC benefit from data dimensionality reduction, with Desik *et al.* (2016)'s identification of relevant data clusters to inform model development of marketing strategies within different insurance product groups proving successful. The Sales and Distribution stage of the IVC uses a similar reduction of dataset features which hold no bearing on insurance customers' likelihood of renewal (Kwak *et al.* 2020).

Knowledge Distillation and Rule Extraction

Knowledge Distillation and Rule Extraction components of AI models refers to the combination of large models to create a smaller, more manageable model (Hinton *et al.* 2015). For instance, both Cheng *et al.* (2020) and Jin *et al.* (2021) investigate optimal insurance strategies (insurance, reinsurance and investment) using the MCAM to develop adequate NN models for their respective prediction tasks. In another work concerning NNs and Knowledge Distillation XAI methods, Kiermayer and Weiß (2021) approximate representative portfolios of both term life insurance plans and Defined Contribution pension plans to aid in determining the insurer's solvency capital requirements. These representative portfolios are inputted in a NN model, which significantly outperforms *k*-means clustering for insurance portfolio grouping and the evaluation of insurers' investment surplus. In another article, Poon (2019) combines simple linear weights and residual components within their model to replicate non-linear effects to resemble a fully parametrised payments-per-claim-incurred model. The combination of models was also utilised by Xu *et al.* (2011) where a random rough subspace method is incorporated into a NN to aid optimised insurance fraud detection.

In terms of extracting actionable knowledge from models, Lee *et al.* (2020) propose a methodology for extracting variables from textual data (word similarities) to use such variables in claims analyses, thus improving actuarial modelling. Similarly, Wang and Xu (2018) apply LDA-based deep learning for the extraction of text features in claims data to detect automobile insurance fraud.

The development of association rules aids in building AI models which are regularly understandable and useful for prediction tasks across the entirety of the IVC. Ravi *et al.* (2017) develop a model for analysing insurance customer complaints and categorising them for insurance customer service offices. Customer grievances are assigned an association rule which are categorised by treating grievance variables as holding a certain degree of membership with the different rules. Association rule learning is also implemented in fraud detection through the identification of frequent fraud occurrence patterns (Verma *et al.* 2017) and the computation of relative weights of variables related to suspicious claim activity using Adaboost AI methods (Viaene *et al.* 2004).

Intrinsically Interpretable Models

Aside from the interpretability techniques outlined above, other researchers have relied on the intrinsic predictive capabilities of models in their research. Through preserving the predictive capabilities of less complex AI models using boosting and optimisation techniques, the predictive power of Intrinsically Interpretable Models proves useful along the IVC.

Researchers implemented Intrinsically Interpretable Models for a range of prediction tasks including: 1) double GLMs to model insurance costs' dispersion and mean (Smyth and Jørgensen 2002), 2) prediction of insurance losses through boosting trees (Guelman 2012), 3) prediction of insurance customers' profitability (Fang *et al.* 2016) and 4) cluster identification and classification (Karamizadeh and Zolfagharifar 2016; Lin *et al.* 2017).

Carfora *et al.* (2019) identified clusters of driver behaviour to inform UBI pricing through unsupervised ML classification techniques and cluster analysis. *K*-means clustering is used to classify driver aggressiveness to inform a risk index of driving behaviour on different road types (primarily urban vs. highway). Benedek and László (2019) compare several interpretable AI techniques in their identification of insurance fraud indicators, which each facilitate the segmentation of such fraud indicators. DTs are highlighted as suitable AI methods for such indicator identification and classification.

4.3 Reproducibility Evaluation

AI, similar to many other research fields, is facing a reproducibility crisis (Hutson 2018). During the systematic review of articles, the author observed the lack of reproducible studies. A summary of the FAIR data principles is provided in *Appendix F*, which informs the reproducibility criteria used in the current systematic review. Following the review of 95 articles, just 13 articles were identified as fully satisfying the criteria of open and reproducible datasets, with 5 articles partially adhering to the FAIR Data guidelines. *Table 8* provides a list of those articles which satisfy the FAIR data principles. This low percentage (~ 19%) of reproducible studies highlights the lack of research cooperation amongst researchers and related institutions, which could hinder further research progress of XAI within the insurance industry.

Table 8: Reproducible research in the systematic articles. Note that the datasets used may be open and accessible for further research, although as this is not explicitly stated in many of the articles and apparent to the author, the result returns '-'.

Reference	Dataset	Findability	Accessibility	Interoperability	Reusability
Alshamsi (2014)	*Allstate Insurance Company*	Yes	Yes	Yes	Yes
Khodairy and Abosamra (2021)	*UAH-DriveSet dataset*	Yes	Yes	Yes	Yes
Shah and Guez (2009)	*Human Mortality Database and National Center for Health Statistics*	Yes	Yes	Yes	Yes
Deprez et al. (2017)	*Swiss Mortality data*	Yes	Yes	Yes	Yes
Gweon et al. (2020)	*190,000 synthetic contracts (extracted from Gan and Valdez (2017))*	Yes	Yes	Yes	Yes
Schelldorfer and Wuthrich (2019)	*French motor third party dataset (678013 observations)*	Yes	Yes	Yes	Yes
Wei and Dan (2019)	*1,200 trading days (Shanghai and Shenzhen 300 Index)*	Yes	Yes	Yes	Yes
Ravi et al. (2017)	*4,032 complaints*	Yes	Yes	Yes	Yes
Herland et al. (2018)	*Centers For Medicare & Medicaid Services*	Yes	Yes	Yes	Yes
Johnson (2019)	*Centers For Medicare & Medicaid Services*	Yes	Yes	Yes	Yes
Lee et al. (2020)	*Wisconsin Local Government Property Insurance Fund dataset (6,030 samples)*	Yes	Yes	Yes	Yes
Smyth and Jorgensen (2002)	*Swedish Third Party Motor Insurance dataset (5,406 claim samples)*	Yes	Yes	Yes	Yes
Xu et al. (2011)	*15,420 claim samples*	Yes	Yes	Yes	Yes
Neumann et al. (2019)	*Aspartus Ltd.*	Yes	-	-	-
Viaene et al. (2002)	*Automobile Insurers Bureau of Massachusetts (1,399 samples)*	Yes	-	-	-
Viaene et al. (2004)	*Automobile Insurers Bureau of Massachusetts (1,400 samples)*	Yes	-	-	-
Benedek and Laszlo (2019)	*Agnoss Knowledge Seeker software (15,420 samples)*	Yes	-	-	Yes
Bian et al. (2018)	*206 samples*	-	-	-	Yes

Several of the articles which did not provide any actionable knowledge on the dataset used were concerned with prediction tasks within the non-life insurance sector (health(care)). Such datasets were either anonymised or did not highlight reasoning for the lacking dataset identification. It is necessary to note that several the reviewed studies utilised a simulated dataset, as real-life datasets can sometimes be too short-term to test the validity of new proposed models[7].

[7] Including, yet not exclusive to, the following reviewed articles: Baudry and Robert (2019), Gan (2013), Gan and Huang (2017) and Kowshalya and Nandhini (2018).

5. Discussion

This paper has thus far provided insight into the presence of XAI applications on the IVC, while also assessing the reviewed articles' potential for reproducibility. Several findings of particular interest from the systematic review are provided in this section which inform later recommendations for further research in the subject area.

5.1 AI's Application on the Insurance Value Chain

The use of AI applications at each stage on the IVC is promising, with a variety of prediction tasks fulfilled by AI applications. In line with Eling *et al.* (2021)'s findings, AI is disrupting the insurance industry in a number of ways. The automation of underwriting tasks and the identification and prevention of fraudulent behaviour are keys areas where AI is impacting the IVC. This is in line with a survey by the Coalition of Insurance Fraud (2020) reporting 56% of insurance companies' surveyed AI as their primary mode of insurance fraud detection. An interesting note is the distinction between Eling *et al.* (2021)'s findings on AI's use in Support Activities and the presence of XAI methods in such activities. The literature search process for this review did not result in any articles concerning XAI use in insurance Support Activities (including HR, IT, Legal and General Management). The author accepts that this finding is likely attributed to restricted keyword searches which do not consider Support Activities, opening the possibility of further research on XAI's presence in insurance companies' Support Activities.

5.2 XAI Definition, Evaluation and Regulatory Compliance

Research on XAI (*Section 2.1*) highlight the disjointed understanding of XAI both across and within industries, thus providing motivation for the current review. There appears no consistent definition of XAI in the reviewed insurance literature, a finding which is in line with Payrovnaziri *et al.* (2020)'s findings of XAI's use and definition in medicine research. The main issue posed by this finding is that the evaluation of XAI methods is made increasingly difficult when there is no

defined definition and scope of XAI. This review aimed to develop an XAI evaluation criteria, incorporating interpretability evaluation as either (i) intrinsic or post-hoc, (ii) local or global and (iii) model-specific or model-agnostic. The results provide an extension to XAI survey research conducted by Adadi and Berrada (2018), Arrieta *et al.* (2020) and Das and Rad (2020) who each defined inter-related taxonomies of XAI. The development of an all-encompassing XAI definition for insurers and AI experts will further the adoption of XAI methods in the insurance industry.

Each definition of XAI found in *Table 2* is derived from the early definition of explainability as the "assignment of causal responsibility" originally cited in Josephson and Josephson (1996). Although each paper providing additional insight into XAI definitions is useful, the lacking cohesion amongst these studies hampers the consolidation of each individual contribution to a widespread accepted XAI definition. The author acknowledges that an all-purpose XAI definition is difficult to determine, as both notions of explainability and interpretability (which are often used interchangeably and used in creating XAI definitions) are domain-specific notions (Freitas 2014; Rudin 2018). In efforts to define XAI within the insurance industry, the author accepts all previous definitions of XAI and findings of XAI use on the IVC to-date and proposes the following XAI definition:

"XAI is the transfer of understanding to AI models' end-users by highlighting key decision-pathways in the model and allowing for human interpretability at various stages of the model's decision process. XAI involves outlining the relationship between model inputs and prediction, meanwhile maintaining predictive accuracy of the model throughout"

In addition to benefitting XAI research, the author notes that a solid definition of XAI pertaining directly to the financial services (the insurance industry) will aid the development of adapted regulation, which is in line with recommendations from Palacio *et al.* (2021). The GDPR (2016) established a regime of "algorithmic accountability" and (insureds) "right to explanation" from decision-making algorithms. XAI promotes such transparent and interpretable traits, yet a comprehensive implementation of these methods necessitates regulatory compliance (Henckaerts *et al.* 2020). In the momentary absence of specific regulation of XAI models, the author highlights the potential for XAI methods to be paired with existing governance measures in the insurance industry to mitigate concerns surrounding the use of novel AI methods until satisfactory regulation

is developed. This recommendation is in line with governance guidelines from (EIOPA 2021), for example the maintenance of human oversight in decision-making processes.

5.3 Research Reproducibility Issue

As outlined in the previous results section, there is a significant lack of reproducible datasets used in AI and insurance research. The lacking emphasis on AI studies' reproducibility in insurance hinders research enhancement within the industry. If researchers used publicly available datasets which satisfy the outlined FAIR Data Principles criteria, their studies' validity can be ascertained. In addition to the validation of research findings, future studies could expand on research methodologies in tandem with advances in technology adoption at each stage on the IVC. The lack of available datasets restricts the furthering of both academic and professional knowledge on the ubiquity of XAI in the insurance industry, limiting researchers' scope for research development.

The author cites the issue of lacking dataset availability in insurance research as a reproducibility crisis, which contrasts with Jensen *et al.* (2021)'s findings which negates a 'replication crisis' in financial economics. Their findings conclude that much of the literature sample can be replicated[8]. This contrast in reproducibility potential in two closely-related fields of finance research shows that reproducibility is possible in insurance research datasets, albeit with additional obstacles in the case of the insurance industry as outlined above.

Although a call for publicly available datasets in various research communities is welcomed (Vetter *et al.* 2016; Christensen and Miguel 2018), the insurance industry is faced with additional challenges regarding dataset access. Data collected by insurance companies is usually of a highly sensitive nature, with insurance proposal forms restricted to asking specific information of proposed insureds in many cases. This is particularly applicable in health and life insurance lines of insurance business. This is supported by the numerous studies within this review which incorporate synthetic datasets in their analysis due to heavily restricted access to insurance companies' datasets. Such an issue is only heightened by recent and ongoing regulation in the

[8] Note that Jensen *et al.* (2021) did not evaluate their data sample using the FAIR Data Principles guidelines, rather a unique Bayesian model of factor replication.

industry which protect proposed insureds' right to privacy and the safe-keeping of sensitive data shared with insurance companies, for instance GDPR regulations applicable to insurance companies operating within the European Union.

Criticisms of AI's trustworthiness and understandability abound, with the insurance industry likely to be used as a benchmark for the satisfactory exchange of sensitive confidential information in AI systems (Rossi 2018; Ferraro 2020). The analysis of current use cases of XAI in insurance is useful to determine the practical use of XAI methods within AI applications on the IVC, but there remains a gap in this analysis. Simply assessing current AI use cases does not inform industry professionals of how they should work *alongside* AI models operating on highly complex insurance-related tasks. Considerations which are particularly applicable to the insurance industry include the high level of trust insurance consumers place in their insurance company and the complex computational tasks associated with the insurance industry (underwriting and pricing of premiums, for instance). This trust allows consumers to provide private and confidential information in health insurance proposal forms, for example. The insurance industry strives to create and use AI models which are mindful of their input data as highly-confidential information in many cases.

The anonymisation of datasets is possible, with such actions already undertaken in genomic research (Corpas *et al.* 2018), government-level policy decisions (Ritchie 2014) and finance trading (Kiyomoto and Fukushima 2018). To create a data anonymisation toolkit for the insurance industry would benefit the research community by enabling research reproducibility and validation, which will in turn allow for the full implementation of XAI models in insurance practices.

6. Conclusion

This systematic review is the first paper to analyse literature on AI applications in the insurance industry and assess the presence of XAI methods in insurance industry practices. The tremendous projected growth of AI applications in insurance, with revenues reaching $3.4 billion by 2024, warrants extensive research in AI and insurance practices. The paper presents an analysis of the reviewed studies' reproducibility in line with the FAIR Data Principles guidelines. The author ignites extensive discussion on XAI suitability to the insurance industry, citing its transparency and understandability to humans as vital to the smooth transition of the industry to an AI-infused industry.

This review assesses 95 articles (comprised of journal articles and conference papers/proceedings) which outline AI applications at each stage of the IVC. The lack of explainability evaluation and consensus on XAI definitions hinders the potential progress of the XAI research field in insurance practices as there is no clear way to evaluate the degree of explainability. This review attempts to bridge this gap by defining XAI criteria in a systematic review of AI in insurance literature. Utilising this XAI criteria the degree of explainability in each AI application is provided, assigning each AI method to a grouped XAI approach, and then evaluating the model's interpretability as: (i) intrinsic or post-hoc, (ii) local or global, and (iii) model-specific or model-agnostic. Findings reiterate the author's hypothesis that XAI methods are popular within insurance research, enabling the transparent use of AI methods in industry research. The transparency XAI methods afford insurance companies enhances the application of AI models in a trust-driven industry.

Additionally, this paper analyses XAI definitions and proposes a revised definition of XAI. This proposed definition is informed by previous XAI definitions in XAI literature and systematic review of literature on AI applications on the IVC. The author acknowledges this definition will not be widely applicable to a wide range of industries, therefore it's reiterated that the proposed XAI definition is applicable to financial services and the insurance industry. This definition will aid in adapting regulation in the insurance industry to suit an AI-rich insurance industry.

Increased focus on reproducible studies is pressurising research communities across a wide range of disciplines, with insurance and risk management proving as no exception to this. The review of

AI methods used on the IVC shows a high degree of explainability in the AI models' operations, whether this was apparent to researchers at the time or not. Further research on the 'FAIRification' process of insurance datasets is recommended to ensure satisfactory data management processes in insurance research is maintained. A framework to safely anonymise insurance data during the research process is highlighted by the author as a remedy for the reproducibility issue in insurance research. The maintenance of insureds' trust in their insurance companies remains of paramount importance to insurers, therefore sensitive data should be anonymised in a rigorous manner to ensure data cannot be traced back to the individual source. Such obstacles in database management in the insurance industry add complexity to the XAI research field in insurance yet is possible to rectify with rigorous database management and FAIR Data guidelines within the industry.

To achieve a substantial understanding of the entire potential of XAI research requires an interdisciplinary effort. The systematic review of XAI methods in different research areas is a stepping-stone to full understanding of the research field, with medicine reviews providing the bulk of knowledge on the topic at the time of writing. Considering the research gap regarding XAI applications along the IVC, this paper is one of the first attempts to provide an overview of XAI's current use within the insurance industry, with intentions of lending further comprehension to the XAI field at large. This paper also provides a pioneering discussion on the relative absence of open and reproducible datasets and research on XAI applications in the insurance industry. A novel framework to aid the production of 'FAIRified' research data in insurance is recommended, which will contribute to enhanced XAI research and understanding within the insurance industry in future years.

7. Appendix

Appendix A: Database of Reviewed Articles

Journal Articles

Reference	Title	Lead Author	Year	Source	Volume	Issue number
Aggour et al. (2006)	Automating the underwriting of insurance applications	Aggour	2006	AI Magazine	27	3
Baecke and Bocca (2017)	The value of vehicle telematics data in insurance risk selection processes	Baecke	2017	Decision Support Systems	98	
Baudry and Robert (2019)	A machine learning approach for individual claims reserving in insurance	Baudry	2019	Applied Stochastic Models in Business and Industry	35	5
Belhadji et al. (2000)	A model for the detection of insurance fraud	Belhadji	2000	The Geneva Papers on Risk and Insurance-Issues and Practice	25	4
Benedek and Laszlo (2019)	Identifying Key Fraud Indicators in the Automobile Insurance Industry Using SQL Server Analysis Services	Benedek	2019	Studia Universitatis Babes-Bolyai	64	2
Bermudez et al. (2008)	A Bayesian dichotomous model with asymmetric link for fraud in insurance	Bermudez	2008	Insurance: Mathematics and Economics	42	2
Boodhun and Jayabalan (2018)	Risk prediction in life insurance industry using supervised learning algorithms	Boodhun	2018	Complex & Intelligent Systems	4	2
Carfora et al. (2019)	A “pay-how-you-drive” car insurance approach through cluster analysis	Carfora	2019	Soft Computing	23	9
Cheng et al. (2011)	Decision making for contractor insurance deductible using the evolutionary support vector machines inference model	Cheng	2011	Expert Systems with Applications	38	6
Cheng et al. (2020)	Optimal insurance strategies: A hybrid deep learning Markov chain approximation approach	Cheng	2020	ASTIN Bulletin: The Journal of the IAA	50	2
Christmann (2004)	An approach to model complex high–dimensional insurance data	Christmann	2004	Allgemeines Statistisches Archiv	88	4
David (2015)	Auto insurance premium calculation using generalized linear models	David	2015	Procedia Economics and Finance	20	
Denuit and Lang (2004)	Non-life rate-making with Bayesian GAMs	Denuit	2004	Insurance: Mathematics and Economics	35	3
Deprez et al. (2017)	Machine learning techniques for mortality modeling	Deprez	2017	European Actuarial Journal	7	2
Desik and Behera (2012)	Acquiring Insurance Customer: The CHAID Way	Desik	2012	IUP Journal of Knowledge Management	10	3
Desik et al. (2016)	Segmentation-Based Predictive Modeling Approach in Insurance Marketing Strategy	Desik	2016	IUP Journal of Business Strategy	13	2
Devriendt et al. (2021)	Sparse regression with multi-type regularized feature modeling	Devriendt	2021	Insurance: Mathematics and Economics	96	
Fang et al. (2016)	Customer profitability forecasting using Big Data analytics: A case study of the insurance industry	Fang	2016	Computers & Industrial Engineering	101	
Frees and Valdez (2008)	Hierarchical insurance claims modeling	Frees	2008	Journal of the American Statistical Association	103	484
Gan (2013)	Application of data clustering and machine learning in variable annuity valuation	Gan	2013	Journal of the American Statistical Association	53	3
Gan and Valdez (2018)	Regression modeling for the valuation of large variable annuity portfolios	Gan	2018	North American Actuarial Journal	22	1
Ghorbani and Farzai (2018)	Fraud detection in automobile insurance using a data mining based approach	Ghorbani	2018	International Journal of Mechatronics, Elektrical and Computer Technology (IJMEC)	8	27

Reference	Title	Lead Author	Year	Source	Volume	Issue number
Guelman (2012)	Gradient boosting trees for auto insurance loss cost modeling and prediction	Guelman	2012	Expert Systems with Applications	39	3
Gweon et al. (2020)	An effective bias-corrected bagging method for the valuation of large variable annuity portfolios	Gweon	2020	ASTIN Bulletin: The Journal of the IAA	50	3
Herland et al. (2018)	The detection of medicare fraud using machine learning methods with excluded provider labels	Herland	2018	Journal of Big Data	5	1
Huang and Meng (2019)	Automobile insurance classification ratemaking based on telematics driving data	Huang	2019	Decision Support Systems	127	
Ibiwoye et al. (2012)	Artificial neural network model for predicting insurance insolvency	Ibiwoye	2012	International Journal of Management and Business Research	2	1
Jain et al. (2019)	Assessing risk in life insurance using ensemble learning	Jain	2019	Journal of Intelligent & Fuzzy Systems	37	2
Jeong et al. (2018)	Association rules for understanding policyholder lapses	Jeong	2018	Risks	6	3
Jiang et al. (2018)	Cost-sensitive parallel learning framework for insurance intelligence operation	Jiang	2018	IEEE Transactions on Industrial Electronics	66	12
Jin et al. (2021)	A hybrid deep learning method for optimal insurance strategies: Algorithms and convergence analysis	Jin	2021	Insurance: Mathematics and Economics	96	
Johnson (2019)	Medicare fraud detection using neural networks	Johnson	2019	Journal of Big Data	6	1
Joram et al. (2017)	A knowledge-based system for life insurance underwriting	Joram	2017	International Journal of Information Technology and Computer Science	3	
Karamizadeh and Zolfagharifar (2016)	Using the clustering algorithms and rule-based of data mining to identify affecting factors in the profit and loss of third party insurance, insurance company auto	Karamizadeh	2016	Indian Journal of science and Technology	9	7
Kašćelan et al. (2016)	A nonparametric data mining approach for risk prediction in car insurance: a case study from the Montenegrin market	Kašćelan	2016	Economic research-Ekonomska istraživanja	29	1
Khodairy and Abosamra (2021)	Driving Behavior Classification Based on Oversampled Signals of Smartphone Embedded Sensors Using an Optimized Stacked-LSTM Neural Networks	Khodairy	2021	IEEE Access	9	
Kiermayer and Weiß (2021)	Grouping of contracts in insurance using neural networks	Kiermayer	2021	Scandinavian Actuarial Journal	2021	4
Kose et al. (2015)	An interactive machine-learning-based electronic fraud and abuse detection system in healthcare insurance	Kose	2015	Applied Soft Computing	36	
Kwak et al. (2020)	Driver Identification Based on Wavelet Transform Using Driving Patterns	Kwak	2020	IEEE Transactions on Industrial Informatics	17	4
Lariviere and van der Poel (2005)	Predicting customer retention and profitability by using random forests and regression forests techniques	Lariviere	2005	Expert systems with applications	29	2
Lee et al. (2020)	Actuarial applications of word embedding models	Lee	2020	ASTIN Bulletin: The Journal of the IAA	50	1
Li et al. (2018)	A principle component analysis-based random forest with the potential nearest neighbor method for automobile insurance fraud identification	Li	2018	Applied Soft Computing	70	
Lin (2009)	Using neural networks as a support tool in the decision making for insurance industry	Lin	2009	Expert Systems with Applications	36	3
Lin et al. (2017)	An ensemble random forest algorithm for insurance big data analysis	Lin	2017	IEEE Access	5	
Liu et al. (2014)	Using multi-class AdaBoost tree for prediction frequency of auto insurance	Liu	2014	Journal of Applied Finance and Banking	4	5
Matloob et al. (2020)	Sequence Mining and Prediction-Based Healthcare Fraud Detection Methodology	Matloob	2020	IEEE Access	8	
Neumann et al. (2019)	Machine Learning-Based Predictions of Customers' Decisions in Car Insurance	Neumann	2019	Applied Artificial Intelligence	33	9
Pathak et al. (2005)	A fuzzy-based algorithm for auditors to detect elements of fraud in settled insurance claims	Pathak	2005	Managerial Auditing Journal	20	6
Poon (2019)	Penalising unexplainability in neural networks for predicting payments per claim incurred	Poon	2019	Risks	7	3
Ravi et al. (2017)	Fuzzy formal concept analysis based opinion mining for CRM in financial services	Ravi	2017	Applied Soft Computing	60	
Sadreddini et al. (2021)	Cancel-for-Any-Reason Insurance Recommendation Using Customer Transaction-Based Clustering	Sadreddini	2021	IEEE Access	9	
Sakthivel and Rajitha (2017)	Artificial intelligence for estimation of future claim frequency in non-life insurance	Sakthivel	2017	Global Journal of Pure and Applied Mathematics	13	6

Reference	Title	Lead Author	Year	Source	Volume	Issue number
Schelldorfer *et al.* (2019)	Nesting classical actuarial models into neural networks	Schelldorfer	2019	Available at SSRN 3320525		
Sevim *et al.* (2016)	Risk Assessment for Accounting Professional Liability Insurance	Sevim	2016	Sosyoekonomi	24	29
Shah and Guez (2009)	Mortality forecasting using neural networks and an application to cause-specific data for insurance purposes	Shah	2009	Journal of Forecasting	28	6
Sheehan *et al.* (2017)	Semi-autonomous vehicle motor insurance: A Bayesian Network risk transfer approach	Sheehan	2017	Transportation Research Part C: Emerging Technologies	82	
Siami *et al.* (2020)	A mobile telematics pattern recognition framework for driving behavior extraction	Siami	2020	IEEE Transactions on Intelligent Transportation Systems	22	3
Smith *et al.* (2000)	An analysis of customer retention and insurance claim patterns using data mining: A case study	Smith	2000	Journal of the Operational Research Society	51	5
Smyth and Jorgensen (2002)	Fitting Tweedie's compound Poisson model to insurance claims data: dispersion modelling	Smyth	2002	ASTIN Bulletin: The Journal of the IAA	32	1
Sun *et al.* (2018)	Abnormal group-based joint medical fraud detection	Sun	2018	IEEE Access	7	
Tillmanns *et al.* (2017)	How to separate the wheat from the chaff: Improved variable selection for new customer acquisition	Tillmanns	2017	Journal of Marketing	81	2
Vaziri and Baheshtina (2016)	A holistic fuzzy approach to create competitive advantage via quality management in services industry (case study: life-insurance services)	Vaziri	2016	Management decision	54	8
Viaene *et al.* (2002)	Auto claim fraud detection using Bayesian learning neural networks	Viaene	2002	Expert Systems with Applications	29	3
Viaene *et al.* (2004)	A case study of applying boosting Naive Bayes to claim fraud diagnosis	Viaene	2004	Journal of Risk and Insurance	69	3
Viaene *et al.* (2005)	A case study of applying boosting Naive Bayes to claim fraud diagnosis	Viaene	2005	IEEE Transactions on Knowledge and Data Engineering	16	5
Wang (2020)	Research on the Features of Car Insurance Data Based on Machine Learning	Wang	2020	Procedia Computer Science	166	
Wang and Xu (2018)	Leveraging deep learning with LDA-based text analytics to detect automobile insurance fraud	Wang	2018	Decision Support Systems	105	
Wei and Dan (2019)	Market fluctuation and agricultural insurance forecasting model based on machine learning algorithm of parameter optimization	Wei	2019	Journal of Intelligent & Fuzzy Systems	37	5
Yan *et al.* (2020)	Research on the UBI Car Insurance Rate Determination Model Based on the CNN-HVSVM Algorithm	Yan	2020	IEEE Access	8	
Yang *et al.* (2006)	Extracting actionable knowledge from decision trees	Yang	2006	IEEE Transactions on Knowledge and data Engineering	19	1
Yang *et al.* (2018)	Insurance premium prediction via gradient tree-boosted Tweedie compound Poisson models	Yang	2018	Journal of Business & Economic Statistics	36	3
Yeo *et al.* (2002)	A mathematical programming approach to optimise insurance premium pricing within a data mining framework	Yeo	2002	Journal of the Operational research Society	53	11

Conference Papers

Reference	Title	Lead Author	Year	Source
Alshami (2014)	Predicting car insurance policies using random forest	Alshami	2014	2014 10th International Conference on Innovations in Information Technology (IIT)
Bian *et al.* (2018)	Good drivers pay less: A study of usage-based vehicle insurance models	Bian	2018	Transportation research part A: policy and practice
Biddle *et al.* (2018)	Transportation research part A: policy and practice	Biddle	2018	Australasian Database Conference
Bonssione *et al.* (2002)	Evolutionary optimization of fuzzy decision systems for automated insurance underwriting	Bonissone	2002	2002 IEEE World Congress on Computational Intelligence. 2002 IEEE International Conference on Fuzzy Systems
Cao and Zhang (2019)	Using PCA to improve the detection of medical insurance fraud in SOFM neural networks	Cao	2019	Proceedings of the 2019 3rd International Conference on Management Engineering, Software Engineering and Service Sciences
Dhieb *et al.* (2019)	Extreme gradient boosting machine learning algorithm for safe auto insurance operations	Dhieb	2019	2019 IEEE International Conference on Vehicular Electronics and Safety (ICVES)
Gan and Huang (2017)	A data mining framework for valuing large portfolios of variable annuities	Gan	2017	Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining
Ghani and Kumar (2011)	Interactive learning for efficiently detecting errors in insurance claims	Ghani	2011	Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining
Kieu *et al.* (2018)	Distinguishing trajectories from different drivers using incompletely labeled trajectories	Kieu	2018	Proceedings of the 27th ACM international conference on information and knowledge management
Kowshalya and Nandhini (2018)	Predicting fraudulent claims in automobile insurance	Kowshalya	2018	2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)
Kumar *et al.* (2010)	Data mining to predict and prevent errors in health insurance claims processing	Kumar	2010	Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining
Kyu and Woraratpanya (2020)	Car Damage Detection and Classification	Kyu	2020	Proceedings of the 11th International Conference on Advances in Information Technology
Lau and Tripathi (2011)	Mine your business—A novel application of association rules for insurance claims analytics	Lau	2011	CAS E-Forum. Arlington: Casualty Actuarial Society
Morik *et al.* (2002)	End-user access to multiple sources-Incorporating knowledge discovery into knowledge management	Morik	2002	International Conference on Practical Aspects of Knowledge Management
Samonte *et al.* (2018)	ICD-9 tagging of clinical notes using topical word embedding	Samonte	2018	Proceedings of the 2018 International Conference on Internet and e-Business
Supraja and Saritha (2017)	Robust fuzzy rule based technique to detect frauds in vehicle insurance	Supraja	2017	2017 International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS)
Tao *et al.* (2012)	Insurance fraud identification research based on fuzzy support vector machine with dual membership	Tao	2012	2012 International Conference on Information Management, Innovation Management and Industrial Engineering
Vassiljeva *et al.* (2017)	Computational intelligence approach for estimation of vehicle insurance risk level	Vassiljeva	2017	2017 International Joint Conference on Neural Networks (IJCNN)
Verma *et al.* (2017)	Fraud detection and frequent pattern matching in insurance claims using data mining techniques	Verma	2017	2017 Tenth International Conference on Contemporary Computing (IC3)
Xu *et al.* (2011)	Random rough subspace based neural network ensemble for insurance fraud detection	Xu	2011	2011 Fourth International Joint Conference on Computational Sciences and Optimization
Yan and Bonissone (2006)	Designing a Neural Network Decision System for Automated Insurance Underwriting	Yan	2006	Insurance Studies
Zahi and Achchab (2019)	Clustering of the population benefiting from health insurance using *k*-means	Zahi	2019	Proceedings of the 4th International Conference on Smart City Applications
Zhang and Kong (2020)	Dynamic estimation model of insurance product recommendation based on Naive Bayesian model	Zhang	2020	Proceedings of the 2020 International Conference on Cyberspace Innovation of Advanced Technologies

Appendix B: The FAIR Data Principles

The below table highlights the main aspects of the FAIR Data Principles guiding framework, as expressly outlined in Wilkinson *et al.* (2016)'s research.

To be Findable:	**F1:** (meta)data are assigned a globally unique and persistent identifier **F2:** data are described with rich metadata (defined by **R1** below) **F3:** metadata clearly and explicitly include the identifier of the data it describes **F4:** (meta)data are registered or indexed in a searchable resource
To be Accessible:	**A1:** (meta)data are retrievable by their identifier using a standardised communications protocol **A1.1:** the protocol is open, free, and universally implementable **A1.2:** the protocol allows for an authentication and authorisation procedure, where necessary **A2:** metadata are accessible, even when the data are no longer available
To be Interoperable:	**I1:** (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation **I2:** (meta)data use vocabularies that follow FAIR principles **I3:** (meta)data include qualified references to other (meta)data
To be Reusable:	**R1:** (meta)data are richly described with a plurality of accurate and relevant attributes **R1.1:** (meta)data are released with a clear and accessible data usage license **R1.2:** (meta)data are associated with detailed provenance **R1.3:** (meta)data meet domain-relevant community standards

**Note:* 'metadata' refers to any case in which the principles should be applied to both metadata and data

Conflicts of Interest: The author declares no conflict of interest.

Abbreviations:

AdaBoost – Adaptive Boosting

AI – Artificial Intelligence

ANN – Artificial Neural Network

BN – Bayesian Network

BPNN – Back Propagation Neural Network

CHAID – Chi-Squared Automatic Interaction Detection

CNN – Convolutional Neural Networks

CPLF – Cost-Sensitive Parallel Learning Framework

CRM – Customer Relationship Management

DFSVM – Dual Membership Fuzzy Support Vector Machine

DL – Deep Learning

ESIM – Evolutionary Support Vector Machine Inference Model

FL – Fuzzy Logic

GAM – Generalised Additive Model

GLM – Generalised Linear Model

HVSVM – Hull Vector Support Vector Machine

IoT – Internet of Things

IVC – Insurance Value Chain

KDD – Knowledge Discovery in Databases

LASSO – Least Absolute Shrinkage and Selection Operator

MCAM – Markov Chain Approximation Method

ML – Machine Learning

NB – Naïve Bayes

NLP – Natural Language Processing

NN – Neural Network

PCA – Principal Component Analysis

RF – Random Forest

SOFM – Self-Organising Feature Map

SOM – Self-Organising Map

UBI – Usage-Based Insurance

WEKA - Waikato Environment for Knowledge Analysis

XAI – Explainable Artificial Intelligence

XGBoost – Extreme Gradient Boosting Algorithms

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